

## **OBJECTIVE QUESTIONS**

**Question 1: What is the total no. of tables present in the data?**

**Solution:**

**Final Answer (if any):**

Total no. of tables in the datasheet=2

**Approach (if any):**

Counted the tables/sheets in the workbook:

- 1.Raw Data
- 2.country description

**Charts & Tables (if any):**

No requirement of creating any chart or table to find the no. tables present in the data

**Question 2: What is the total no. of attributes present in the data?**

**Solution:**

**Final Answer (if any):**

Total number of attributes in the Raw Data table = 20

(If the lookup table "country description" is also included, total attributes across both tables = 22.)

**Approach (if any):**

Counted the header columns in the Raw Data sheet from A to T, which gives 20 columns.

The country description sheet is a lookup/reference table, so it is not included in the main attribute count.

**Charts & Tables (if any):**

-No requirement of creating any chart or table

### Question 3: How many categorical columns are there in the data? [Search about categorical and continuous data, and try to answer this question]

#### Solution:

##### Final Answer (if any):

There are **16 categorical columns** in the dataset.

##### Approach (if any):

To solve this question, I first understood the difference between **categorical data** and **continuous data**.

**Categorical data** refers to columns that contain **labels, names, groups, classes, or yes/no type values**, whereas **continuous data** contains measurable numeric values such as ratings, average cost, votes, latitude, and longitude.

After reviewing the restaurant dataset, I checked each column one by one and classified it based on the type of values it stores.

Columns such as **Restaurant ID, Restaurant Name, Country Code, Country, City, Address, Locality, Locality Verbose, Cuisines, Currency, Has Table Booking, Has Online Delivery, Is Delivering Now, Switch to Order Menu, Rating text, and Rating color** were treated as categorical because they represent identifiers, text labels, categories, or fixed-value groups rather than measurable quantities.

On the other hand, columns like **Average Cost for two, Price range, Aggregate rating, Votes, Latitude, and Longitude** were treated as continuous or numeric fields because they are used for measurement and comparison.

Based on this classification, the total number of **categorical columns is 16**.

These columns are especially important in analysis because they are mainly used for **filtering, grouping, segmentation, pivot tables, and dashboard slicers**.

##### Charts & Tables (if any):

**Question 4: The data consists of some inconsistent and missing values so ensure that the data used for further analysis is cleaned.**

**Solution:**

**Final Answer (if any):**

The dataset was cleaned before starting the analysis to make sure the results are accurate and reliable. First, all blank or missing values were replaced with **“Unknown”** so that empty cells do not affect filtering, sorting, or pivot table analysis. Next, the **Date column** was converted into the proper date format to ensure correct date-based analysis and consistency across the dataset. In addition, some **wrong spellings and inconsistent text values** in the data were identified and corrected so that similar entries are grouped properly during analysis. This cleaning process improved the overall data quality and made the dataset ready for further analysis such as ratings, cuisines, country-wise trends, and expansion decisions.

**Approach (if any):**

1. Checked the dataset for **blank or missing values** in important columns.
2. Replaced all blank cells with **“Unknown”** to avoid empty data issues.
3. Identified the **Date column** and changed it into the **proper date format**.
4. Reviewed text-based columns for **spelling mistakes and inconsistent naming**.
5. Corrected the wrong spellings so similar values are treated as the same category.
6. Ensured the cleaned data is consistent and ready for **pivot table analysis and dashboard creation**.

**Charts & Tables (if any):**

No Chart required

**Question 5: Using the LookUp functions, fill up the countries in the original data using the country code.**

**Solution:**

**Final Answer (if any):**

The country names were successfully filled in the original dataset by using the VLOOKUP function to map each Country Code with its corresponding Country Name from the country description sheet.

#### Approach (if any):

The **Country Code** available in the original dataset was matched with the **Country Code** in the **country description** sheet. Using the **VLOOKUP** function, the corresponding **Country Name** was retrieved and filled into the original data.

#### Steps followed:

1. Selected the first cell in the **Country Name** column.
2. Applied the formula:

=VLOOKUP(C2,'country description'!A:B,2,FALSE)

3. Pressed **Enter**.
4. Dragged the formula down to apply it to all rows.

This ensured that all country codes were converted into their respective country names.

#### Charts & Tables (if any):

- Table / Chart Name:

country description

- Description:

A lookup reference table containing country code and country name, used to populate the country names in the original dataset.

## Question 6: Create a table to represent the number of restaurants opened in each country.

### Solution:

#### Final Answer (if any):

Created a country-wise summary table showing the number of restaurants in each country using the Country column.

#### Approach (if any):

Used the Pivot Table:

Selected the cleaned dataset

Go to Insert → Pivot table

In Rows: added **Country**

In Values: added **RestaurantID**

Set summary as **COUNTA**

This will return the number of restaurants for each country.

**Charts & Tables (if any):**

| <b><i>Country Name</i></b> | <b>COUNTA of RestaurantID</b> |
|----------------------------|-------------------------------|
| Australia                  | 24                            |
| Brazil                     | 60                            |
| Canada                     | 4                             |
| India                      | 8652                          |
| Indonesia                  | 21                            |
| New Zealand                | 40                            |
| Philippines                | 22                            |
| Qatar                      | 20                            |
| Singapore                  | 20                            |
| South Africa               | 60                            |
| Sri Lanka                  | 20                            |
| Turkey                     | 34                            |
| United Arab Emirates       | 60                            |
| United Kingdom             | 80                            |
| United States of America   | 434                           |
| <b>Grand Total</b>         | <b>9551</b>                   |

**Question 7: Also, the management wants to look at the number of restaurants opened each year, so provide them with something here.**

## **Solution:**

### **Final Answer (if any):**

Created a year-wise summary of the number of restaurants opened by extracting the year from the Datekey\_Opening column and then counting the number of restaurants for each year.

### **Approach (if any):**

Used a Pivot Table after extracting the year:

First created a new column named `Opening_Year`

Used `YEAR(T2)`

Drag the formula down

Then created Pivot Table:

Select the cleaned dataset

Go to Insert → Pivot table

In Rows: add `Opening_Year`

In Values: add `RestaurantID`

Set summary as `COUNTA`

This will show how many restaurants were opened each year.

### **Charts & Tables (if any):**

| <i>Opening_Year</i> | COUNTA of RestaurantID |
|---------------------|------------------------|
| 2010                | 1080                   |
| 2011                | 1098                   |
| 2012                | 1022                   |
| 2013                | 1061                   |
| 2014                | 1051                   |
| 2015                | 1024                   |
| 2016                | 1027                   |
| 2017                | 1086                   |
| 2018                | 1102                   |
| <b>Grand Total</b>  | <b>9551</b>            |

## Question 8: What is the total number of restaurants in India in the price range of 4?

### Solution:

#### Final Answer (if any):

The total number of restaurants in India with price range=4 is 388

#### Approach (if any):

The total number of restaurants was calculated using the COUNTIFS function by applying two conditions:

- 1.Country Name=India
- 2.Price Range=4

#### Formula used:

=COUNTIFS('Cleaned Data'!M:M,"India",'Cleaned Data'!R:R,4)

This formula counts all restaurants in the Cleaned Data sheet where the country is India and the price range is 4.

#### Charts & Tables (if any):

- Table / Chart Name:  
<Name of the table or chart used>
- Description:  
<What the table or chart represents>

## Question 9: What is the average number of voters for the restaurants in each country according to the data?

### Solution:

#### Final Answer (if any):

Create a country-wise table showing the average number of voters (Votes) for restaurants in each country.

#### Approach (if any):

Use a Pivot Table:

Select the cleaned dataset

Go to Insert → Pivot table

In Rows: add Country

In Values: add Votes

Change summary from Sum to Average

This will calculate the average number of voters for restaurants in each country.

#### Charts & Tables (if any):



| <i>Country Name (Objective Question 5)</i> | <b>AVERAGE of Votes</b> |
|--------------------------------------------|-------------------------|
| Australia                                  | 111.4166667             |
| Brazil                                     | 19.61666667             |
| Canada                                     | 103                     |
| India                                      | 137.212552              |
| Indonesia                                  | 772.0952381             |
| New Zealand                                | 243.025                 |
| Philippines                                | 407.4090909             |
| Qatar                                      | 163.8                   |
| Singapore                                  | 31.9                    |
| South Africa                               | 315.1666667             |
| Sri Lanka                                  | 146.45                  |
| Turkey                                     | 431.4705882             |
| United Arab Emirates                       | 493.5166667             |
| United Kingdom                             | 205.4875                |
| United States of America                   | 428.2211982             |
| <b>Grand Total</b>                         | <b>156.9097477</b>      |

**Question 10: Calculate the average rating for all the restaurants that have price\_range < 4 and provide online delivery. Use only the “IF” function, Logical Operators, and Aggregation functions to solve this problem. [Note: Don’t use Conditional aggregation in this question.]**

## **Solution:**

### **Final Answer (if any):**

The average rating was calculated using a **helper column** with the **IF** function and then applying the **AVERAGE** function.

**Helper column formula:** =IF(AND(M2="Yes",P2<4),S2,"")

**Final formula:** =AVERAGE('Cleaned Data'!X2:X)

### **Approach (if any):**

A helper column was created using the **IF** function with the **AND** logical operator to return the **Rating** only when:

**Price Range < 4**

**Online Delivery = Yes**

Then, the **AVERAGE** function was applied to the helper column to calculate the required average rating.

**Charts & Tables (if any):**

- Table / Chart Name:  
<Name of the table or chart used>
- Description:  
<What the table or chart represents>

**Question 11: Using Conditional formatting highlight the rows of restaurants that are located in the countries or cities that you've suggested to the management for opening new restaurants.**

**Solution:**

**Final Answer (if any):**

Conditional Formatting was applied to highlight restaurant rows that match the profile of **high-performing premium restaurants**, which can be used as a benchmark for identifying strong expansion opportunities in the suggested countries/cities.

The rows were highlighted using the following conditions:

**Price\_range = 4**

**Rating >= 4.9**

**Average\_Cost\_for\_two between 1100 and 1250**

These conditions helped identify restaurants that represent:

premium market positioning

excellent customer satisfaction

strong customer spending capacity

Such restaurants can be considered as reference points while evaluating locations for opening new restaurants.

### Approach (if any):

To support the management in identifying ideal expansion zones, I used **Conditional Formatting** to highlight rows that match the characteristics of **top-performing premium restaurants**.

### Logic used for highlighting rows:

I selected restaurants that satisfy all of the following conditions:

#### Price Range = 4

This represents restaurants in the **premium / luxury segment**

It helps identify markets that can support high-end restaurant concepts

#### Rating >= 4.9

This indicates **excellent customer satisfaction**

Restaurants with such high ratings are strong indicators of quality and market acceptance

#### Average Cost for Two between 1100 and 1250

This range reflects a **high-spending customer segment**

It helps identify locations where customers are comfortable paying premium prices

### Why this approach was used:

Instead of highlighting all rows manually, Conditional Formatting was used to automatically identify restaurants that match the **ideal expansion benchmark**.

This approach helps in:

quickly spotting high-value restaurant records

identifying premium market clusters

understanding the type of locations that support upscale restaurants

supporting data-driven expansion decisions

### Excel logic used (conceptually):

Rows were highlighted when all three conditions were true:

Price\_range = 4

Rating >= 4.9

Average\_Cost\_for\_two >= 1100

Average\_Cost\_for\_two <= 1250

This can be applied using a formula-based conditional formatting rule across the entire dataset table.

**Charts & Tables (if any):**

- Table / Chart Name:  
<Name of the table or chart used>
- Description:  
<What the table or chart represents>

**Question 12: Create a new customized price column that consists of the abbreviation/symbol of the currency along with the Average\_cost\_for\_two value. [Use string operations to do this task]**

**Solution:**

**Final Answer (if any):**

A new customized price column was created by first mapping each restaurant's country to its corresponding currency reference from the **Country Description** sheet, and then using **VLOOKUP** with the **Average\_Cost\_for\_two** field to generate a country-specific price value.

The logic used was:

=VLOOKUP(C2,'country description'!\$A\$2:\$C\$16,3,0)\*R2

Where:

**C2** = CountryCode

**'country description'!\$A\$2:\$C\$16** = lookup range containing CountryCode, Country Name, and Currency reference

**3** = fetches the currency-related value from the 3rd column

**R2** = Average\_Cost\_for\_two

This formula produced a customized price output based on the restaurant's country and average cost.

### Approach (if any):

To create the customized price column, I followed a two-step approach:

#### Step 1: Create a currency reference table

In the **Country Description** sheet, I created a supporting table containing:

**Country Code**

**Country Name**

**Currency** (country-specific currency reference value)

This table was used as the base lookup table for mapping each restaurant to its respective currency information.

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#### Step 2: Use VLOOKUP with Average Cost for Two

I used the following formula:

`=VLOOKUP(C2,'country description'!$A$2:$C$16,3,0)*R2`

#### Formula explanation:

`VLOOKUP(C2, . . . )` looks up the **CountryCode**

It fetches the corresponding **currency value** from the 3rd column of the **Country Description** table

Then it multiplies that value by:

- **R2 = Average\_Cost\_for\_two**

This creates a **country-adjusted customized price value** based on the restaurant's local currency context.

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### **Business logic behind this approach:**

Restaurants belong to different countries

Each country uses a different currency system

So instead of keeping only one standard cost value, the formula helps create a **country-specific customized price representation**

This makes the data more useful for comparing restaurant costs in local market context

### **Charts & Tables (if any):**

- Table / Chart Name:  
<Name of the table or chart used>
- Description:  
<What the table or chart represents>

**Question 13: How can you create an array formula in Excel or Google Sheets to count the number of restaurants listed that do not offer online delivery, are in the lowest price range, and have an average cost for two people less than or equal to 250 Indian Rupees?**

### **Solution:**

#### **Final Answer (if any):**

An **array formula** was used to count the number of restaurants that satisfy all of the following conditions:

- **Online Delivery = No**
- **Price Range = 1** (lowest price range)
- **Average Cost for Two  $\leq$  250**

#### **Google Sheets Array Formula:**

```
=SUM(ARRAYFORMULA(IF((P2:P9551="No")*(R2:R9551=1)*(J2:J9551<=250),1,0))  
)
```

Where:

- **P** = Online Delivery
- **R** = Price Range
- **J** = Average\_cost\_for\_two

This formula returns the **total count of restaurants** matching all required conditions.

### **Approach (if any):**

An **array formula** was created using **ARRAYFORMULA**, **IF**, and logical multiplication of conditions.

### **Logic used:**

Each row was checked for:

1. **Online Delivery = "No"**
2. **Price Range = 1**
3. **Average Cost for Two  $\leq$  250**

If all three conditions were true, the formula returned **1**; otherwise, it returned **0**.  
Finally, all 1's were summed to get the required count.

### **Charts & Tables (if any):**

- Table / Chart Name:  
<Name of the table or chart used>
- Description:  
<What the table or chart represents>

## **SUBJECTIVE QUESTIONS**

**Question 1: Suggest a few countries where the team can open newer restaurants with lesser competition. Which visualization/technique will you use here to justify the suggestions?**

**Solution:**

**Approach:**

We need to suggest few countries where the team can open newer restaurants with lesser competition . For that we will create a pivot table in which we will considering two factors first will be count of restaurant and second will be average of rating.

On these two factors we can suggest few countries.

The less should be the no. of restaurant and the more should be the rating.

Two factors:

- 1.count of restaurant in that country(it should be less in no.)
- 2.average of rating(it should be high)

**Charts & Tables:**

- Table / Chart Name:

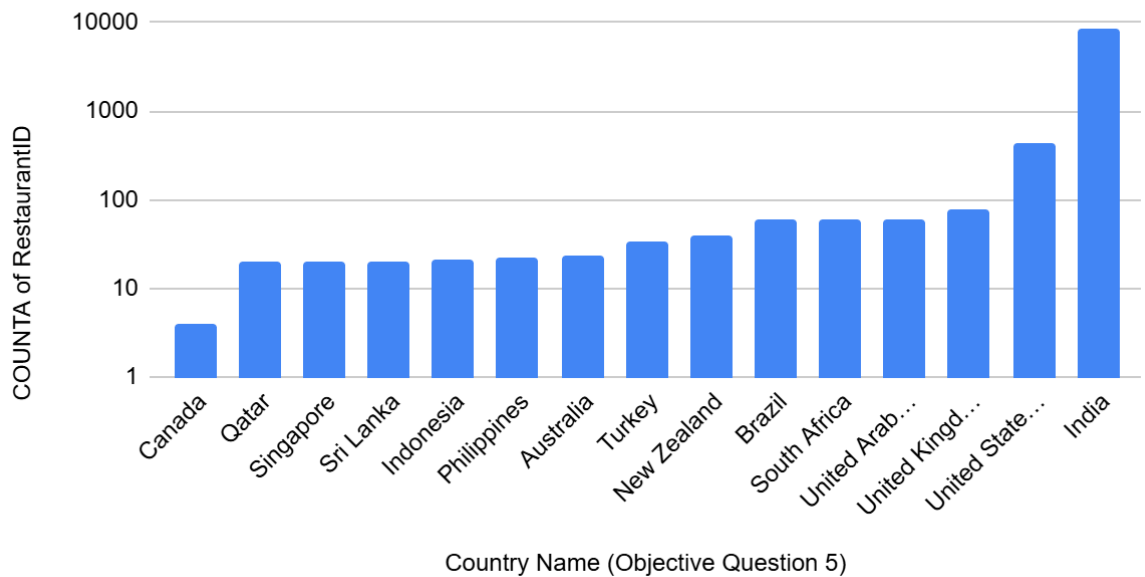
| Country   |    |             |
|-----------|----|-------------|
| Canada    | 4  | 3.575       |
| Qatar     | 20 | <b>4.06</b> |
| Singapore | 20 | 3.575       |
| Sri Lanka | 20 | 3.87        |



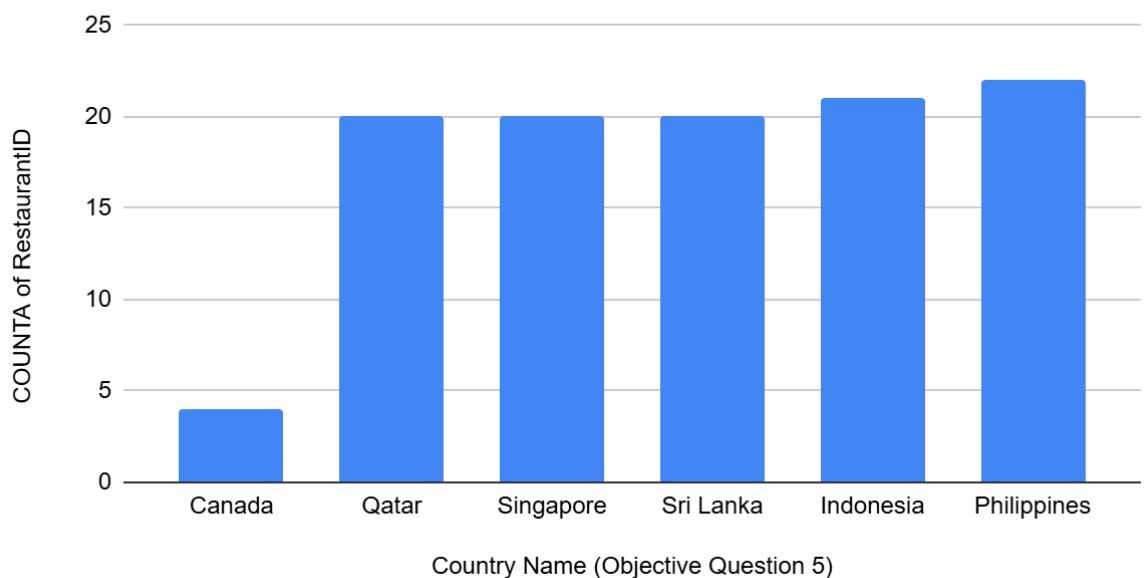
|                          |      |             |
|--------------------------|------|-------------|
| Indonesia                | 21   | 4.295238095 |
| Philippines              | 22   | 4.468181818 |
| Australia                | 24   | 3.658333333 |
| Turkey                   | 34   | 4.3         |
| New Zealand              | 40   | 4.2625      |
| Brazil                   | 60   | 3.846666667 |
| South Africa             | 60   | 4.21        |
| United Arab Emirates     | 60   | 4.233333333 |
| United Kingdom           | 80   | 4.1         |
| United States of America | 434  | 4.011290323 |
| India                    | 8652 | 2.770550162 |

|                   |
|-------------------|
| Suggested Country |
| Canada            |
| Qatar             |
| Singapore         |
| Sri Lanka         |
| Indonesia         |
| Philippines       |

COUNTA of RestaurantID vs. Country Name (Objective Question 5)



COUNTA of RestaurantID vs. Country Name (Objective Question 5)



- Description:

This table represent the country name along with no. of restaurant and average of rating

### Insights:

Countries Like India has very high no. of restaurants. While countries like Canada has very low no. no. of restaurants.

Having good rating means the high acceptance of customer

So, we should go with those countries which have low no. of restaurants and good rating.

### Recommendations:

Based on the analysis, the team should consider opening newer restaurants in countries with **less competition and relatively better ratings**, such as:

Canada  
Qatar  
Singapore  
Sri Lanka  
Philippines  
Indonesia

These countries appear to offer:

- lower competitive pressure
- better market entry opportunities
- potential for expansion with less saturation

## Question 2: Come up with the names of States and cities in the suggested countries suitable for opening restaurants.

### Solution:

#### Approach:

Within the **suggested countries**, the objective was to identify **cities** suitable for opening new restaurants.

To do this, a **Pivot Table** was created with:

- **Rows:** Country, City
- **Values:**
  - **Count of Restaurant ID**
  - **Average of Rating**
  - **Average of Average\_cost\_for\_two** with Currency Symbol

The cities were shortlisted based on the following criteria:

1. **Lower number of restaurants** → indicates **less competition**

2. **Higher average rating** → indicates **better customer satisfaction / market acceptance**
3. **Reasonable or premium average cost for two** → helps understand spending capacity and positioning potential

The final suggested cities were selected by balancing:

- **low competition**
- **strong ratings**
- **acceptable market value**

### Charts & Tables:

- Table / Chart Name:

| Country     | City                    | COUNTA of RestaurantID | AVERAGE of Rating | AVERAGE of Average_cost_f or_two with currency symbol (Objective question 12) |
|-------------|-------------------------|------------------------|-------------------|-------------------------------------------------------------------------------|
| Canada      | Chatham-Kent            | 1                      | 3.7               | ₹1,525.00                                                                     |
|             | <b>Consort</b>          | <b>1</b>               | 3                 | ₹1,525.00                                                                     |
|             | <b>Vineland Station</b> | <b>1</b>               | 4.3               | ₹4,270.00                                                                     |
|             | Yorkton                 | 1                      | 3.3               | ₹1,525.00                                                                     |
| Indonesia   | Bandung                 | 1                      | 4.2               | ₹795.00                                                                       |
|             | Bogor                   | 2                      | 3.85              | ₹848.00                                                                       |
|             | Jakarta                 | 16                     | 4.35625           | ₹1,634.72                                                                     |
|             | <b>Tangerang</b>        | <b>2</b>               | 4.3               | ₹1,325.00                                                                     |
| Philippines | Makati City             | 2                      | 4.65              | ₹1,702.00                                                                     |
|             | <b>Mandaluyong City</b> | <b>4</b>               | 4.625             | ₹2,960.00                                                                     |
|             | Pasay City              | 3                      | 4.366666667       | ₹4,933.33                                                                     |
|             | <b>Pasig City</b>       | <b>3</b>               | 4.633333333       | ₹1,381.33                                                                     |
|             | Quezon City             | 1                      | 4.8               | ₹1,184.00                                                                     |
|             | <b>San Juan City</b>    | <b>2</b>               | 4.25              | ₹1,258.00                                                                     |
|             | Santa Rosa              | 2                      | 3.8               | ₹1,221.00                                                                     |
|             | Tagaytay City           | 1                      | 4.5               | ₹1,776.00                                                                     |

|           |             |    |       |           |
|-----------|-------------|----|-------|-----------|
|           | Taguig City | 4  | 4.525 | ₹2,553.00 |
| Qatar     | Doha        | 20 | 4.06  | ₹5,101.50 |
| Singapore | Singapore   | 20 | 3.575 | ₹9,578.63 |
| Sri Lanka | Colombo     | 20 | 3.87  | ₹665.00   |

|                  |
|------------------|
| Suggested Cities |
| Vineland Station |
| Jakarta          |
| Tangerang        |
| Makai City       |
| Mandaluyong City |
| Pasay City       |
| Pasig City       |
| Quezon           |
| San Juan City    |
| Tagaytay City    |
| Taguig City      |
| Doha             |
| Singapore        |

A Pivot Table having **Countries and Cities in rows**, and the following measures in values:

**COUNTA of RestaurantID**

**AVERAGE of Rating**

**AVERAGE of Average\_cost\_for\_two with currency symbol (Objective Question 12)**

This table was used to identify cities in the suggested countries that have:

fewer restaurants (lower competition)

good customer ratings

suitable average spending levels

## Insights:

### 1. Canada

Cities like **Vineland Station**, **Chatham-Kent**, **Consort**, and **Yorkton** each have only **1 restaurant**, which means **very low competition**.

Among them:

- **Vineland Station** has the **highest rating (4.3)**, making it the strongest option.
- **Chatham-Kent** also performs reasonably with **3.7 rating**.

**Yorkton** and **Consort** have lower ratings, so they are less attractive.

### 2. Indonesia

**Jakarta** has the highest number of restaurants (**16**), so competition is relatively higher.

However, **Jakarta** also has a **strong average rating (4.36)** and a balanced average cost, showing a strong market.

**Tangerang** has only **2 restaurants** and a **high rating (4.3)**, making it a very attractive low-competition option.

**Bandung** also has only **1 restaurant** and a **good rating (4.2)**, so it is another strong candidate.

**Bogor** has lower rating compared to the other cities.

### 3. Philippines

Several cities have **low restaurant count (1–4)** and **excellent ratings**, making this country highly attractive.

Best-performing cities include:

- **Quezon City** → **1 restaurant, 4.8 rating**
- **Makati City** → **2 restaurants, 4.65 rating**
- **Pasig City** → **3 restaurants, 4.63 rating**
- **Mandaluyong City** → **4 restaurants, 4.63 rating**
- **Taguig City** → **4 restaurants, 4.53 rating**
- **Tagaytay City** → **1 restaurant, 4.5 rating**
- **Pasay City** → **3 restaurants, 4.37 rating**

This shows **strong demand with relatively low competition**.

#### 4. Qatar

**Doha** has **20 restaurants**, so competition is not very low.

But it has a **good rating (4.06)** and **high average cost for two**, which suggests a **premium market opportunity**.

Suitable if the team wants a **high-value / premium positioning** rather than only low competition.

#### 5. Singapore

**Singapore** has **20 restaurants** and a lower rating (**3.58**) compared to many other shortlisted cities.

Also, average cost is very high.

This market may be more competitive and less attractive than the better-rated cities in Philippines or Indonesia.

#### 6. Sri Lanka

**Colombo** has **20 restaurants**, which means competition is higher than the low-count cities.

Rating is **3.87**, which is decent but not as strong as top options in Philippines or Indonesia.

Could be a secondary option.

#### Recommendations:

Based on the city-level analysis, the most suitable cities for opening new restaurants are those with **lower competition (fewer restaurants)** and **higher ratings**.

#### Final Recommended Cities:

##### Canada

**Vineland Station** (*Best option in Canada – 1 restaurant, 4.3 rating*)

**Chatham-Kent** (*Secondary option – 1 restaurant, 3.7 rating*)

##### Indonesia

**Tangerang** (2 restaurants, 4.3 rating – strong low-competition option)

**Bandung** (1 restaurant, 4.2 rating – very good option)

**Jakarta** (Higher competition, but strong rating and large market potential)

**Philippines (Strongest country for expansion in this analysis)**

**Quezon City** (1 restaurant, 4.8 rating – top recommendation)

**Makati City** (2 restaurants, 4.65 rating)

**Pasig City** (3 restaurants, 4.63 rating)

**Mandaluyong City** (4 restaurants, 4.63 rating)

**Taguig City** (4 restaurants, 4.53 rating)

**Tagaytay City** (1 restaurant, 4.5 rating)

**Pasay City** (3 restaurants, 4.37 rating)

**Qatar**

**Doha** (Good premium market option, though competition is higher)

**Conditional / Secondary Recommendations**

**Singapore** (Only if targeting a premium/high-spending market, but not ideal for low competition)

**Colombo** (Moderate option, not as strong as Philippines/Indonesia cities)

**Question 3: According to the countries you suggested, what is the current quality regarding ratings for restaurants that are open there?**

**Solution:**

**Approach:**



To evaluate the current quality of restaurants in the shortlisted countries, I analyzed the **average rating of restaurants that are currently open**.

### Methodology used:

Filtered the dataset to include only the **suggested countries** from the previous question.

Considered only **open/active restaurants** for the analysis.

Created a **Pivot Table** in Excel to calculate the **average rating** for each country.

Compared the country-wise average ratings to identify which countries currently have better restaurant quality.

### Assumption:

Higher **average rating** indicates **better restaurant quality** and **higher customer satisfaction**.

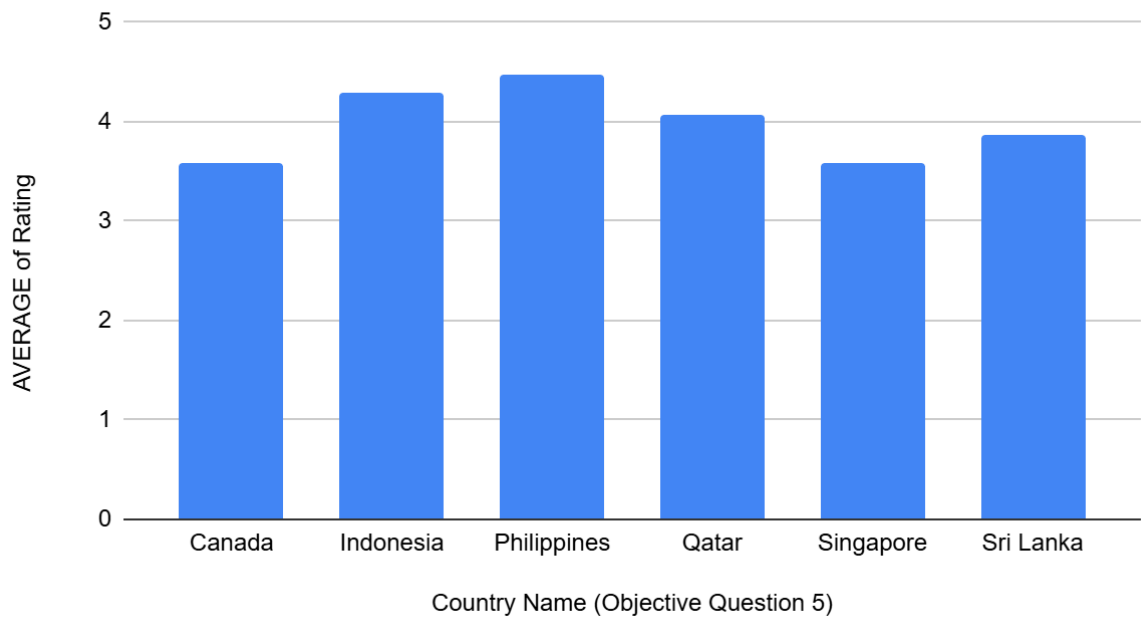
Countries with ratings **above 4.0** are considered strong quality markets for expansion.

### Charts & Tables:

- Table / Chart Name:

| Answer | Country     | AVERAGE of Rating |
|--------|-------------|-------------------|
|        | Canada      | 3.575             |
|        | Indonesia   | 4.295238095       |
|        | Philippines | 4.468181818       |
|        | Qatar       | 4.06              |
|        | Singapore   | 3.575             |
|        | Sri Lanka   | 3.87              |
|        | Grand Total | 4.045794393       |

### AVERAGE of Rating vs. Country Name (Objective Question 5)



#### - Description:

This pivot table shows the **average rating of open restaurants** in each shortlisted country.

It helps compare the **current quality level** of restaurant operations across the selected markets.

#### Table Values Used:

Canada → **3.575**

Indonesia → **4.295**

Philippines → **4.468**

Qatar → **4.060**

Singapore → **3.575**

Sri Lanka → **3.870**

Grand Total → **4.046**

□ Insert your pivot table screenshot here in the Excel subsheet for this section.

### Insights:

**Philippines** has the **highest average rating (4.468)**, which indicates the **best current restaurant quality** among all selected countries.

**Indonesia (4.295)** also shows a **very strong performance**, meaning restaurants there are highly rated by customers.

**Qatar (4.060)** is another strong market, with an average rating above **4.0**, showing **very good quality**.

**Sri Lanka (3.870)** falls in the **good quality** range, but it is slightly lower than the top three countries.

**Canada (3.575)** and **Singapore (3.575)** have the **lowest average ratings** among the shortlisted countries, which suggests comparatively moderate restaurant quality.

The **overall average rating (4.046)** indicates that the selected countries generally have a **good to very good restaurant quality level**.

### Recommendations:

**Prioritize expansion in Philippines**, as it has the **highest average rating** and reflects the **strongest customer satisfaction**.

**Indonesia** should also be considered a **top expansion market** because of its high restaurant quality and positive customer perception.

**Qatar** can be treated as a **stable and promising market** due to its rating being above 4.0.

**Sri Lanka** can be considered as a **secondary opportunity**, especially if the business wants to explore moderate but positive markets.

**Canada and Singapore** should be approached more carefully, as their lower ratings suggest stronger competition, inconsistent quality, or the need for a more differentiated market entry strategy.

**Question 4: Also, what is the current expenditure on food in the suggested countries, so we can keep our financial expenditure in control?**

**Solution:**

### Approach:

To understand the **current expenditure on food** in the suggested countries, I analyzed the **average cost for two people** across restaurants in each shortlisted country.

## Methodology used:

Filtered the dataset to include only the **suggested countries** identified in the previous analysis.

Used the column “**Average\_cost\_for\_two with currency symbol (Objective question 12)**” as the main financial indicator.

Created a **Pivot Table** in Excel to calculate the **average food expenditure for two people** for each country.

Compared the country-wise expenditure to identify:

**high-cost markets**

**moderate-cost markets**

**budget-friendly markets**

## Assumption:

**Average cost for two** is treated as the best indicator of **current customer spending on food**.

Countries with **lower average cost for two** are more suitable if the goal is to **control operational and financial expenditure**.

Countries with **higher average cost for two** may offer premium pricing opportunities, but they can also increase cost pressure and budget requirements.

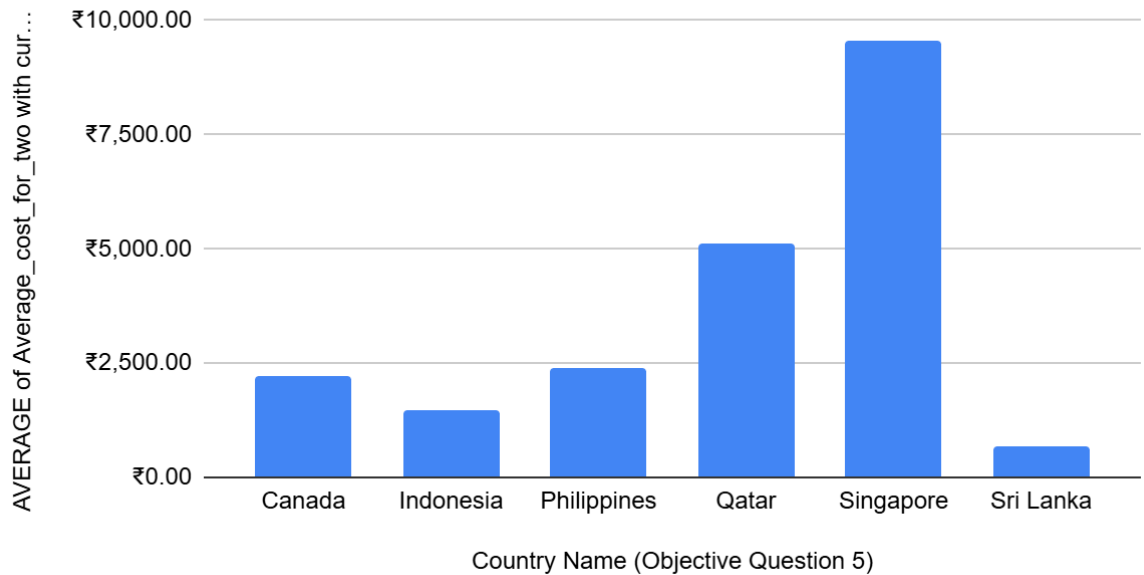
## Charts & Tables:

- Table / Chart Name:

| Answer | Country     | AVERAGE of Average_cost_for_two with currency symbol (Objective question 12) |
|--------|-------------|------------------------------------------------------------------------------|
|        | Canada      | ₹2,211.25                                                                    |
|        | Indonesia   | ₹1,490.31                                                                    |
|        | Philippines | ₹2,378.09                                                                    |
|        | Qatar       | ₹5,101.50                                                                    |

|  |                    |                  |
|--|--------------------|------------------|
|  | <b>Singapore</b>   | <b>₹9,578.63</b> |
|  | Sri Lanka          | ₹665.00          |
|  | <b>Grand Total</b> | <b>₹3,732.36</b> |

AVERAGE of Average\_cost\_for\_two with currency symbol  
(Objective question 12) vs. Country Name (Objective Questio...



- Description:

This pivot table shows the **average expenditure on food for two people** in each suggested country.

It helps compare the **current spending level** across markets so that expansion decisions can be aligned with **budget control and financial planning**.

### Table Values Used:

Canada → **₹2,211.25**

Indonesia → **₹1,490.31**

Philippines → **₹2,378.09**

Qatar → **₹5,101.50**

Singapore → **₹9,578.63**

Sri Lanka → **₹665.00**

Grand Total → **₹3,732.36**

### Insights:

**Singapore** has the **highest average food expenditure** at **₹9,578.63**, which makes it the **most expensive market** among the shortlisted countries.

**Qatar** is the **second most expensive market** with an average cost of **₹5,101.50**, indicating a relatively high spending environment.

**Philippines (₹2,378.09)** and **Canada (₹2,211.25)** fall in the **moderate expenditure range**, making them financially more manageable than Singapore and Qatar.

**Indonesia (₹1,490.31)** shows a **lower and more controlled spending level**, which can support cost-efficient expansion.

**Sri Lanka (₹665.00)** has the **lowest average food expenditure**, making it the **most budget-friendly market** among all the selected countries.

The **overall average expenditure** across all selected countries is **₹3,732.36**, but this is significantly influenced by the very high costs in **Singapore and Qatar**.

### Recommendations:

If the main goal is to **keep financial expenditure under control**, the best countries to prioritize are:

**Sri Lanka**

**Indonesia**

**Canada / Philippines** (moderate-cost options)

**Sri Lanka** should be considered a **highly cost-efficient market** because it has the **lowest average food expenditure**, which can help reduce pricing pressure and operational cost sensitivity.

**Indonesia** is also a **strong cost-controlled option**, as it combines:

**low average expenditure**

and from previous analysis, **high restaurant quality (high average rating)**

**Philippines** can be a **balanced expansion choice**, because although its food expenditure is moderate, it also had the **highest restaurant rating** in Question 3, which makes it attractive from both quality and affordability perspectives.

**Singapore** should be approached carefully if budget control is important, because its **very high food expenditure** may lead to:

higher customer pricing expectations

higher market entry cost

premium positioning requirements

**Qatar** can be considered only if the business is targeting a **premium segment**, as the expenditure level is relatively high.

**Question 5: Come up with the names of restaurants from the recommended states that are our biggest competitors and also those that are rated in the lower brackets, i.e. 1-2 or 2-3.**

## **Solution:**

### **Approach:**

To answer this question, I used the restaurant data from the **recommended countries/cities** identified in the earlier expansion analysis and divided the restaurants into two categories:

#### **1. Biggest Competitors**

These are restaurants in the recommended markets that have **very high customer ratings (4.6–4.9)**.

High-rated restaurants were considered the **strongest competitors**, because they already have strong customer satisfaction, brand acceptance, and market presence.

#### **2. Lower Bracket Restaurants**

These are restaurants that fall into the **lower rating ranges**:

**1–2**

**2–3**

These restaurants were identified to understand **weak performers** and spot areas where a better-managed restaurant could capture market share.

## **Methodology Used:**

Filtered the recommended market data for restaurants in:

**Indonesia**

**Philippines**

**Qatar**

Selected restaurants with **high ratings (4.6–4.9)** as **biggest competitors**

Used the corrected low-rating dataset to identify restaurants with:

**Rating = 1 to 2**

**Rating = 2 to 3**

Grouped them location-wise to compare strong competitors vs weak performers

This approach helps management understand:

**who the strongest existing competitors are**

**where poor-performing restaurants exist**

**where market gaps can be exploited with a better restaurant concept**

**Charts & Tables:**

- Table / Chart Name:

| Country     | City                    | RestaurantName                           | AVERAGE of Rating |
|-------------|-------------------------|------------------------------------------|-------------------|
| Indonesia   | Jakarta                 | Satoo - Hotel Shangri-La                 | 4.6               |
|             |                         | Sushi Masa                               | 4.9               |
|             |                         | <b>Talaga Sampireun</b>                  | 4.9               |
|             |                         | <b>Toodz House</b>                       | 4.6               |
|             |                         | Union Deli                               | 4.6               |
|             | <b>Tangerang</b>        | <b>Talaga Sampireun</b>                  | 4.9               |
| Philippines | Makati City             | Le Petit Souffle                         | 4.8               |
|             | <b>Mandaluyong City</b> | <b>Ooma</b>                              | 4.9               |
|             |                         | Sambo Kojin                              | 4.8               |
|             | Pasay City              | Spiral - Sofitel Philippine Plaza Manila | 4.9               |



|       |                   |                           |     |
|-------|-------------------|---------------------------|-----|
|       | <b>Pasig City</b> | <b>Locavore</b>           | 4.8 |
|       |                   | Silantro Fil-Mex          | 4.9 |
|       | Quezon City       | Silantro Fil-Mex          | 4.8 |
|       | Taguig City       | NIU by Vikings            | 4.7 |
| Qatar | Doha              | Gymkhana                  | 4.7 |
|       |                   | Mainland China Restaurant | 4.9 |
|       |                   | Zaffran Dining Experience | 4.6 |

| Country   | City           | RestaurantName        | AVERAGE of Rating |  |
|-----------|----------------|-----------------------|-------------------|--|
| Australia | Mayfield       | Star Buffet           | 2.9               |  |
|           | Montville      | Poets Cafe            | 2.4               |  |
|           | Paynesville    | Pier 70               | 2.6               |  |
| Brazil    | Brasilia       | Chez Michou           | 3                 |  |
|           |                | Sandubas Caf          | 1                 |  |
|           | Rio de Janeiro | Quiosque Chopp Brahma | 1                 |  |
|           | S  o Paulo     | Cantinho da Gula      | 1                 |  |
|           |                | Divino Fog  o         | 1                 |  |
|           |                | Super Grill           | 1                 |  |
| Canada    | Consort        | Consort Restaurant    | 3                 |  |

### Insights:

The recommended markets **Indonesia, Philippines, and Qatar** contain several highly rated restaurants, showing strong restaurant demand and good expansion potential.

Restaurants like **Sushi Masa, Talaga Sampireun, Ooma, Spiral - Sofitel Philippine Plaza Manila, and Mainland China Restaurant** are the biggest competitors with ratings between **4.6 and 4.9**.

**Indonesia (Jakarta and Tangerang)** shows strong premium competition, indicating customers value quality dining experiences.

The **Philippines** has multiple high-performing cities, which suggests a wider and more mature food market.

**Doha (Qatar)** also has strong premium-rated competitors, making it a high-potential but competitive market.

The presence of many highly rated restaurants means these markets already support restaurants with excellent service and food quality.

On the other hand, the lower-bracket dataset shows many restaurants with ratings between **1 and 3**, especially in **Faridabad**.

A large number of low-rated restaurants indicates poor customer satisfaction, weak service, or inconsistent food quality in some markets.

This creates a clear opportunity for a new restaurant to capture customers by offering better quality, hygiene, service, and overall experience.

Overall, the data shows that while strong competitors exist in the recommended markets, there are also visible market gaps that a well-managed restaurant can successfully target.

### **Recommendations:**

#### **1) Benchmark Against High-Rated Competitors**

The company should study the strongest competitors in recommended markets, especially:

**Sushi Masa**

**Talaga Sampireun**

**Ooma**

**Spiral - Sofitel Philippine Plaza Manila**

**Silantro Fil-Mex**

**Mainland China Restaurant**

These restaurants can be used as **benchmark examples** for:

menu quality

service standards

customer experience

premium pricing strategy

---

## **2) Prioritize Expansion in Markets with Strong Demand**

Based on competitor quality, the most promising expansion markets remain:

**Jakarta / Tangerang (Indonesia)**

**Makati / Mandaluyong / Pasay / Pasig / Taguig (Philippines)**

**Doha (Qatar)**

These locations show:

strong demand for quality restaurants

customer willingness to pay for better experiences

established dining culture

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## **3) Use Low-Rated Restaurants to Identify Market Gaps**

Low-rated restaurants indicate where customers may be dissatisfied.

This creates an opportunity to enter the market with:

better food quality

faster service

improved ambience

stronger hygiene standards

online delivery and table booking options

This is especially useful in markets where many restaurants are rated between **1 and 3**.

---

## **4) Avoid Copying Weak Competitor Models**

The company should avoid business practices seen in low-rated restaurants, such as:

poor service quality

inconsistent food taste

poor cleanliness

weak online presence

lack of customer engagement

## Question 6: Which cuisines should we focus on in the newer restaurants to get better feedback? Does the choice of cuisines affect the restaurant ratings?

### Solution:

#### Approach:

To answer this question, I created a pivot table using the **Cuisines** column and calculated the **Average of Rating** for each cuisine type.

Then I sorted the cuisines from highest to lowest rating to identify which cuisine categories receive better customer feedback.

This helped in comparing both single cuisines and multi-cuisine combinations.

The goal was to understand whether cuisine preference influences customer ratings.

Based on the ratings, higher-performing cuisines were considered more suitable for new restaurant planning.

#### Charts & Tables:

- Table / Chart Name:

| Answer | Cuisines                   | AVERAGE of Rating |
|--------|----------------------------|-------------------|
|        | African                    | 4.7               |
|        | American                   | 4.566666667       |
|        | American, Asian, Burger    | 4.6               |
|        | American, Asian, European, | 4.6               |

|  |                                             |            |
|--|---------------------------------------------|------------|
|  | Seafood                                     |            |
|  | American,<br>Asian,<br>Italian,<br>Seafood  | 4.5        |
|  | American, Bar<br>Food                       | 4.5        |
|  | American,<br>BBQ,<br>Sandwich               | 4.9        |
|  | American,<br>Breakfast,<br>Burger           | 4.5        |
|  | American,<br>Burger                         | 4.5        |
|  | American,<br>Burger,<br>Cafe                | 4.7        |
|  | American,<br>Burger,<br>Fast<br>Food        | 4.6        |
|  | American,<br>Burger,<br>Grill               | 4.9        |
|  | <b>American,<br/>Caribbean,<br/>Seafood</b> | <b>4.9</b> |

|  |                                             |      |
|--|---------------------------------------------|------|
|  | Americ<br>an,<br>Coffee<br>and<br>Tea       | 4.9  |
|  | Americ<br>an,<br>Contine<br>ntal,<br>Burger | 4.6  |
|  | Americ<br>an,<br>Dessert<br>s               | 4.65 |
|  | Americ<br>an,<br>Dessert<br>s,<br>Steak     | 4.7  |
|  | Americ<br>an,<br>Hawaii<br>an               | 4.5  |
|  | Americ<br>an,<br>Japane<br>se,<br>Sushi     | 4.6  |
|  | Americ<br>an,<br>Pizza,<br>Bar<br>Food      | 4.7  |
|  | Americ<br>an,<br>Sandwi<br>ch,<br>Salad     | 4.7  |
|  | Americ<br>an,<br>Sandwi<br>ch, Tea          | 4.9  |
|  | Americ<br>an,<br>Seafoo<br>d,<br>Cajun      | 4.6  |

|  |                                                    |                 |
|--|----------------------------------------------------|-----------------|
|  | Americ<br>an,<br>Southe<br>rn                      | 4.5             |
|  | Americ<br>an,<br>Southe<br>rn,<br>Southw<br>estern | 4.5             |
|  | Argenti<br>ne,<br>Americ<br>an                     | 4.5             |
|  | Asian                                              | 4.7             |
|  | Asian,<br>Burmes<br>e                              | 4.6             |
|  | Asian,<br>Chines<br>e                              | 4.6             |
|  | Asian,<br>Chines<br>e, Thai,<br>Japane<br>se       | 4.7             |
|  | Asian,<br>Contine<br>ntal,<br>Fast<br>Food         | 4.5             |
|  | Asian,<br>Indone<br>sian,<br>Wester<br>n           | 4.6             |
|  | Asian,<br>Sushi,<br>Seafoo<br>d                    | 4.5             |
|  | Asian,<br>Thai                                     | 4.6             |
|  | Asian,<br>Vietna<br>mese                           | 4.5             |
|  | Bakery,<br>Dessert<br>s                            | 4.7333<br>33333 |

|  |                                         |      |
|--|-----------------------------------------|------|
|  | Bar Food                                | 4.9  |
|  | BBQ, Breakfast, Southern                | 4.9  |
|  | BBQ, Burger, Desserts                   | 4.5  |
|  | BBQ, Caribbean, Cajun                   | 4.5  |
|  | BBQ, Caribbean, Seafood                 | 4.6  |
|  | Beverages, Bar Food, Fast Food          | 4.5  |
|  | Beverages, International                | 4.8  |
|  | Biryani, North Indian, Chinese, Mughlai | 4.6  |
|  | Brazilian                               | 4.65 |
|  | Brazilian, Bar Food                     | 4.8  |
|  | Brazilian, Bar Food, Beverages          | 4.6  |



|  |                                                         |      |
|--|---------------------------------------------------------|------|
|  | Brazilian,<br>Healthy Food,<br>Juices,<br>Pizza         | 4.6  |
|  | Brazilian,<br>Steak                                     | 4.6  |
|  | Breakfast,<br>Mexican                                   | 4.5  |
|  | Breakfast,<br>Sandwich                                  | 4.5  |
|  | British                                                 | 4.9  |
|  | British,<br>Contemporary                                | 4.5  |
|  | Burger                                                  | 4.8  |
|  | Burger,<br>American,<br>Fast Food,<br>Italian,<br>Pizza | 4.5  |
|  | Burger,<br>Bar Food,<br>Steak                           | 4.9  |
|  | Burger,<br>Desserts, Bar Food                           | 4.5  |
|  | Burger,<br>Fast Food                                    | 4.5  |
|  | Cafe                                                    | 4.6  |
|  | Cafe,<br>American                                       | 4.6  |
|  | Cafe,<br>Bakery,                                        | 4.55 |

|  |                                                        |     |
|--|--------------------------------------------------------|-----|
|  | Desserts                                               |     |
|  | Cafe, Burger                                           | 4.8 |
|  | Cafe, Charcoal Grill, Steak                            | 4.5 |
|  | Cafe, Continental                                      | 4.6 |
|  | Cafe, Continental, Chinese, Italian                    | 4.6 |
|  | Cafe, Continental, Fast Food                           | 4.6 |
|  | Cafe, Continental, Italian, Mexican, Chinese, American | 4.7 |
|  | Cafe, Fast Food, Street Food                           | 4.7 |
|  | Cafe, Italian                                          | 4.6 |
|  | Cafe, Italian, Coffee and Tea, Western, Indonesian     | 4.6 |

|  |                                                                                                            |      |
|--|------------------------------------------------------------------------------------------------------------|------|
|  | Cafe,<br>Italian,<br>Contine<br>ntal                                                                       | 4.6  |
|  | Cafe,<br>Korean<br>,<br>Dessert<br>s                                                                       | 4.5  |
|  | Cafe,<br>North<br>Indian,<br>Chines<br>e                                                                   | 4.9  |
|  | Cafe,<br>Tapas,<br>South<br>African,<br>Bevera<br>ges,<br>Healthy<br>Food,<br>Dessert<br>s,<br>Spanis<br>h | 4.5  |
|  | Cajun                                                                                                      | 4.7  |
|  | Chines<br>e                                                                                                | 4.9  |
|  | Chines<br>e, Dim<br>Sum                                                                                    | 4.75 |
|  | Chines<br>e,<br>Japane<br>se,<br>Sushi                                                                     | 4.5  |
|  | Chines<br>e,<br>Pizza                                                                                      | 4.6  |
|  | Chines<br>e,<br>Steak                                                                                      | 4.6  |
|  | Chines<br>e, Thai                                                                                          | 4.5  |
|  | Conte<br>mporar<br>y                                                                                       | 4.5  |

|  |                                                                                                   |     |
|--|---------------------------------------------------------------------------------------------------|-----|
|  | Conte<br>mporar<br>y,<br>Italian                                                                  | 4.8 |
|  | Contine<br>ntal,<br>Americ<br>an                                                                  | 4.5 |
|  | Contine<br>ntal,<br>Americ<br>an,<br>Italian,<br>North<br>Indian,<br>Chines<br>e, Cafe            | 4.7 |
|  | Contine<br>ntal,<br>Cafe,<br>Spanis<br>h,<br>Italian,<br>Europe<br>an,<br>Greek,<br>Mediterranean | 4.5 |
|  | Contine<br>ntal,<br>Goan,<br>Seafoo<br>d,<br>North<br>Indian                                      | 4.6 |
|  | Contine<br>ntal,<br>Indian                                                                        | 4.9 |
|  | Contine<br>ntal,<br>Italian,<br>Asian,<br>Indian                                                  | 4.7 |
|  | Contine<br>ntal,<br>Italian,<br>Pizza,<br>Asian                                                   | 4.6 |

|  |                                                                                                                                       |      |
|--|---------------------------------------------------------------------------------------------------------------------------------------|------|
|  | Contine<br>ntal,<br>Kerala                                                                                                            | 4.5  |
|  | Contine<br>ntal,<br>Mexica<br>n                                                                                                       | 4.6  |
|  | Contine<br>ntal,<br>Middle<br>Eastern<br>, Asian                                                                                      | 4.8  |
|  | Contine<br>ntal,<br>North<br>Indian,<br>Chines<br>e                                                                                   | 4.8  |
|  | Contine<br>ntal,<br>South<br>African,<br>Bevera<br>ges,<br>Dessert<br>s,<br>Seafoo<br>d, Grill,<br>Ice<br>Cream,<br>Internat<br>ional | 4.5  |
|  | Dessert<br>s                                                                                                                          | 4.75 |
|  | Dessert<br>s,<br>Bakery,<br>Wester<br>n                                                                                               | 4.6  |
|  | Dessert<br>s, Bar<br>Food                                                                                                             | 4.7  |
|  | Dessert<br>s,<br>Bí_rek                                                                                                               | 4.7  |
|  | Dessert<br>s, Cafe                                                                                                                    | 4.8  |

|  |                                                 |             |
|--|-------------------------------------------------|-------------|
|  | Desserts, Cafe, Mexican                         | 4.7         |
|  | Desserts, Ice Cream                             | 4.533333333 |
|  | Desserts, Sandwich, Ice Cream                   | 4.6         |
|  | European                                        | 4.7         |
|  | European, Asian, Indian                         | 4.9         |
|  | European, Contemporary                          | 4.9         |
|  | European, Desserts                              | 4.8         |
|  | European, German                                | 4.9         |
|  | European, Mediterranean, North Indian           | 4.8         |
|  | European, North Indian, Mediterranean, American | 4.5         |
|  | European,                                       | 4.6         |

|  |                                                                                    |                 |
|--|------------------------------------------------------------------------------------|-----------------|
|  | South African                                                                      |                 |
|  | European,<br>South African,<br>Steak                                               | 4.5             |
|  | Fast Food                                                                          | 4.7333<br>33333 |
|  | Filipino                                                                           | 4.65            |
|  | Filipino,<br>Japanese,<br>Asian                                                    | 4.5             |
|  | Filipino,<br>Mexican                                                               | 4.85            |
|  | Finger Food,<br>American,<br>Continental,<br>North Indian,<br>Italian              | 4.6             |
|  | Finger Food,<br>North Indian,<br>Italian,<br>Continental,<br>Thai,<br>South Indian | 4.5             |
|  | French                                                                             | 4.8             |
|  | French,<br>Brazilian,<br>Beverages                                                 | 4.6             |
|  | French,<br>Japanese,<br>Desserts                                                   | 4.8             |

|  |                                                |                 |
|--|------------------------------------------------|-----------------|
|  | German                                         | 4.8             |
|  | Goan,<br>Seafood,<br>North<br>Indian           | 4.5             |
|  | Greek                                          | 4.7             |
|  | Hawaiian                                       | 4.5             |
|  | Hawaiian,<br>Seafood                           | 4.9             |
|  | Ice<br>Cream                                   | 4.62            |
|  | Ice<br>Cream,<br>Desserts                      | 4.7             |
|  | Ice<br>Cream,<br>Desserts,<br>Continental      | 4.5             |
|  | Indian                                         | 4.76            |
|  | Indian,<br>Cafe,<br>Curry                      | 4.5             |
|  | Indian,<br>Continental                         | 4.8             |
|  | Indian,<br>Curry                               | 4.5             |
|  | Indian,<br>Mediterranean,<br>Middle<br>Eastern | 4.6             |
|  | Indian,<br>North<br>Indian                     | 4.7333<br>33333 |
|  | Indian,<br>North<br>Indian,                    | 4.7             |



|  |                                                          |                 |
|--|----------------------------------------------------------|-----------------|
|  | Curry,<br>Cafe                                           |                 |
|  | Indian,<br>Pakista<br>ni,<br>Curry                       | 4.7             |
|  | Indian,<br>Street<br>Food                                | 4.5             |
|  | Internat<br>ional                                        | 4.9             |
|  | Internat<br>ional,<br>Indian                             | 4.5             |
|  | Internat<br>ional,<br>Italian,<br>Southe<br>rn           | 4.8             |
|  | Internat<br>ional,<br>Mediter<br>ranean,<br>Sandwi<br>ch | 4.5             |
|  | Internat<br>ional,<br>Tapas,<br>Vegeta<br>rian           | 4.5             |
|  | Italian                                                  | 4.5285<br>71429 |
|  | Italian,<br>Americ<br>an,<br>Pizza                       | 4.8             |
|  | Italian,<br>Bakery,<br>Contine<br>ntal                   | 4.9             |
|  | Italian,<br>Contine<br>ntal,<br>Europe<br>an,<br>Cafe    | 4.55            |

|  |                                                            |                 |
|--|------------------------------------------------------------|-----------------|
|  | Italian,<br>Deli                                           | 4.9             |
|  | Italian,<br>French,<br>Cafe                                | 4.5             |
|  | Italian,<br>Pizza                                          | 4.5333<br>33333 |
|  | Italian,<br>Pizza,<br>Sandwi<br>ch                         | 4.6             |
|  | Japane<br>se                                               | 4.6666<br>66667 |
|  | Japane<br>se,<br>Korean                                    | 4.8             |
|  | Japane<br>se,<br>Sushi                                     | 4.7666<br>66667 |
|  | Juices,<br>Dessert<br>s                                    | 4.5             |
|  | Juices,<br>Healthy<br>Food                                 | 4.7             |
|  | Kebab                                                      | 4.55            |
|  | Kebab,<br>Turkish<br>Pizza                                 | 4.6             |
|  | Latin<br>Americ<br>an                                      | 4.6             |
|  | Latin<br>Americ<br>an,<br>Mexica<br>n,<br>Southw<br>estern | 4.6             |
|  | Mediterranean                                              | 4.5             |
|  | Mediterranean,<br>Asian,<br>Contine<br>ntal,<br>North      | 4.6             |

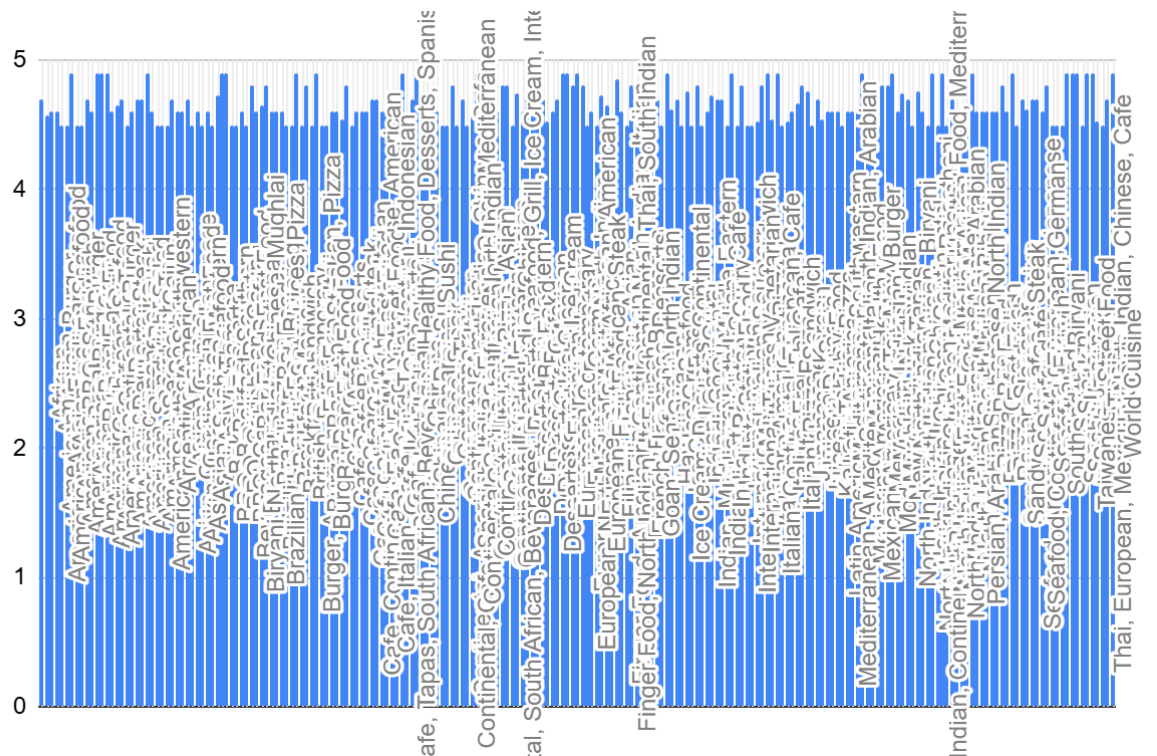
|  |                                             |     |
|--|---------------------------------------------|-----|
|  | Indian,<br>Arabian                          |     |
|  | Mediterranean,<br>Italian                   | 4.6 |
|  | Mexican                                     | 4.6 |
|  | Mexican,<br>American,<br>Healthy Food       | 4.9 |
|  | Mexican,<br>American,<br>Tex-Mex            | 4.5 |
|  | Mexican,<br>American,<br>Tex-Mex,<br>Burger | 4.7 |
|  | Mexican,<br>Vegetarian                      | 4.5 |
|  | Modern Indian                               | 4.8 |
|  | Modern Indian,<br>North Indian              | 4.8 |
|  | Mughlai,<br>Lucknowi                        | 4.9 |
|  | New American,<br>Tapas                      | 4.6 |

|  |                                                                                    |                 |
|--|------------------------------------------------------------------------------------|-----------------|
|  | North Indian                                                                       | 4.7428<br>57143 |
|  | North Indian, American, Asian, Biryani                                             | 4.7             |
|  | North Indian, Asian                                                                | 4.5             |
|  | North Indian, Chinese                                                              | 4.7666<br>66667 |
|  | North Indian, Chinese, Continental                                                 | 4.6             |
|  | North Indian, Chinese, Continental, Rajasthani                                     | 4.5             |
|  | North Indian, Chinese, Mediterranean                                               | 4.9             |
|  | North Indian, Continental, Beverages, Italian, Burger, Healthy Food, Mediterranean | 4.5             |

|  |                                             |       |
|--|---------------------------------------------|-------|
|  | North Indian, European, Mediterranean       | 4.9   |
|  | North Indian, Fast Food, Italian, Asian     | 4.5   |
|  | North Indian, Italian, Chinese, Mexican     | 4.7   |
|  | North Indian, Mediterranean, Asian, Arabian | 4.6   |
|  | North Indian, Mughlai                       | 4.525 |
|  | North Indian, South Indian, Bihari          | 4.5   |
|  | North Indian, South Indian, Mughlai         | 4.9   |
|  | Persian , Arabian , Lebanese, North Indian  | 4.6   |
|  | Pizza                                       | 4.6   |

|  |                                                                 |      |
|--|-----------------------------------------------------------------|------|
|  | Pizza,<br>Bar<br>Food                                           | 4.6  |
|  | Pizza,<br>Cafe,<br>Italian                                      | 4.6  |
|  | Pizza,<br>Fast<br>Food                                          | 4.6  |
|  | Pizza,<br>Sandwi<br>ch                                          | 4.8  |
|  | Ramen<br>,<br>Japane<br>se                                      | 4.6  |
|  | Sandwi<br>ch,<br>Seafoo<br>d,<br>Cajun                          | 4.9  |
|  | Sandwi<br>ch,<br>Seafoo<br>d,<br>Steak                          | 4.5  |
|  | Scottis<br>h, Cafe                                              | 4.7  |
|  | Seafoo<br>d                                                     | 4.62 |
|  | Seafoo<br>d,<br>Americ<br>an,<br>Mediterranean,<br>Japane<br>se | 4.7  |
|  | Seafoo<br>d,<br>Contine<br>ntal,<br>Europe<br>an,<br>Germa<br>n | 4.7  |

|  |                                                                        |                 |
|--|------------------------------------------------------------------------|-----------------|
|  | Seafood,<br>French                                                     | 4.6             |
|  | Seafood, Kiwi                                                          | 4.8             |
|  | South<br>Indian                                                        | 4.5             |
|  | South<br>Indian,<br>Biryani                                            | 4.5             |
|  | Southern                                                               | 4.5             |
|  | Steak                                                                  | 4.9             |
|  | Sunda,<br>Indonesian                                                   | 4.9             |
|  | Sushi,<br>Japanese                                                     | 4.9             |
|  | Sushi,<br>Teriyaki                                                     | 4.5             |
|  | Taiwanese,<br>Street<br>Food                                           | 4.9             |
|  | Tapas                                                                  | 4.9             |
|  | Thai                                                                   | 4.5333<br>33333 |
|  | Thai,<br>European,<br>Mexican,<br>North<br>Indian,<br>Chinese,<br>Cafe | 4.5             |
|  | Unknown                                                                | 4.7             |
|  | World<br>Cuisine                                                       | 4.9             |



#### - Description:

This pivot table shows the average rating of different cuisines and cuisine combinations.

It helps identify which cuisines are most preferred by customers based on feedback. The table can also be supported with a bar chart to highlight top-rated cuisines more clearly.

This makes it easier to select suitable cuisine types for future expansion.

#### Insights:

The analysis shows that **cuisine choice does affect restaurant ratings**.

Some cuisines and cuisine combinations have clearly higher average ratings than others.

Premium, specialty, and multi-cuisine options generally perform better in customer feedback.

Cuisines like **Chinese, Japanese/Sushi, Continental combinations, International, and Steak-based options** appear among the top-rated groups.

This suggests that customers respond more positively to cuisines that offer quality, variety, or a unique dining experience.

#### Recommendations:

For newer restaurants, the team should focus on **high-rated cuisines or strong multi-cuisine combinations** that are already performing well in the data.

It is better to offer cuisines that are both popular and capable of giving a premium or differentiated experience.



The final cuisine mix should also be selected according to the target country/city and customer preferences.

Management can use the top-rated cuisines as a benchmark while planning menus for expansion.

Overall, cuisine strategy should be treated as an important factor because it directly influences customer feedback.

## Question 7: According to our current data, should we go for online delivery and table booking? Does that affect the customer's ratings?

### Solution:

#### Approach:

To determine whether the business should adopt **online delivery** and **table booking**, I analyzed how these two service features affect **customer ratings**.

#### Methodology used:

Divided the restaurants into two categories for each feature:

Has\_Online\_Delivery → Yes / No

Has\_Table\_Booking → Yes / No

Created separate **Pivot Tables** in Excel to calculate the **average customer rating** for each category.

Compared the average ratings of restaurants that **offer** these services versus those that **do not offer** them.

The objective was to check whether these features are associated with **higher customer satisfaction**, measured through ratings.

#### Assumption:

A **higher average rating** indicates **better customer experience and satisfaction**.

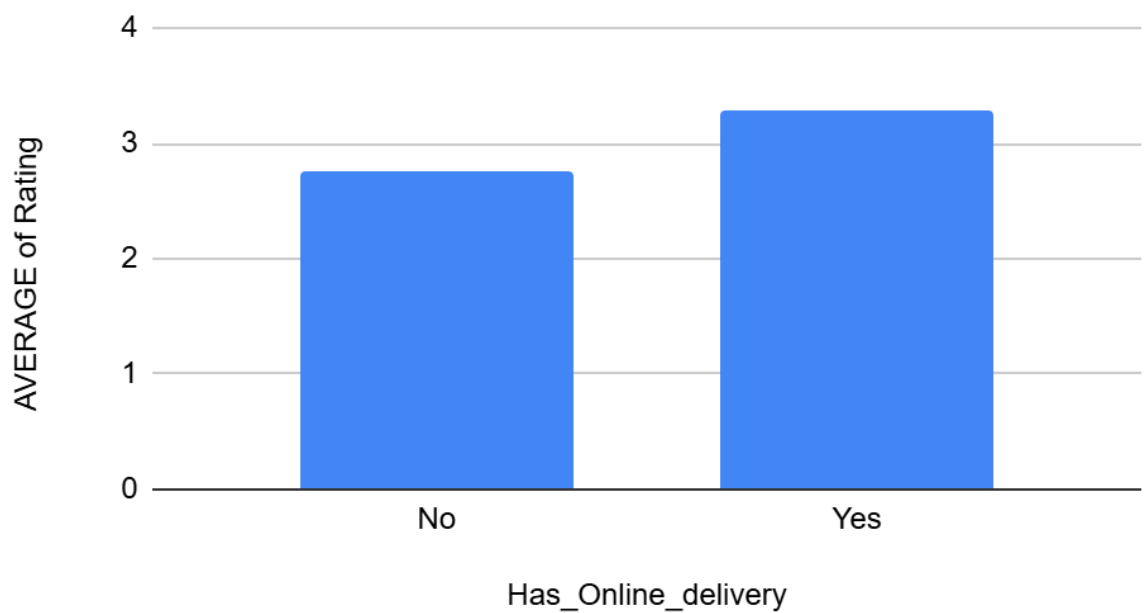
If restaurants with online delivery or table booking have significantly higher ratings, then these services likely create a **positive impact on customer perception** and should be considered in the expansion strategy.

### Charts & Tables:

- Table / Chart Name:

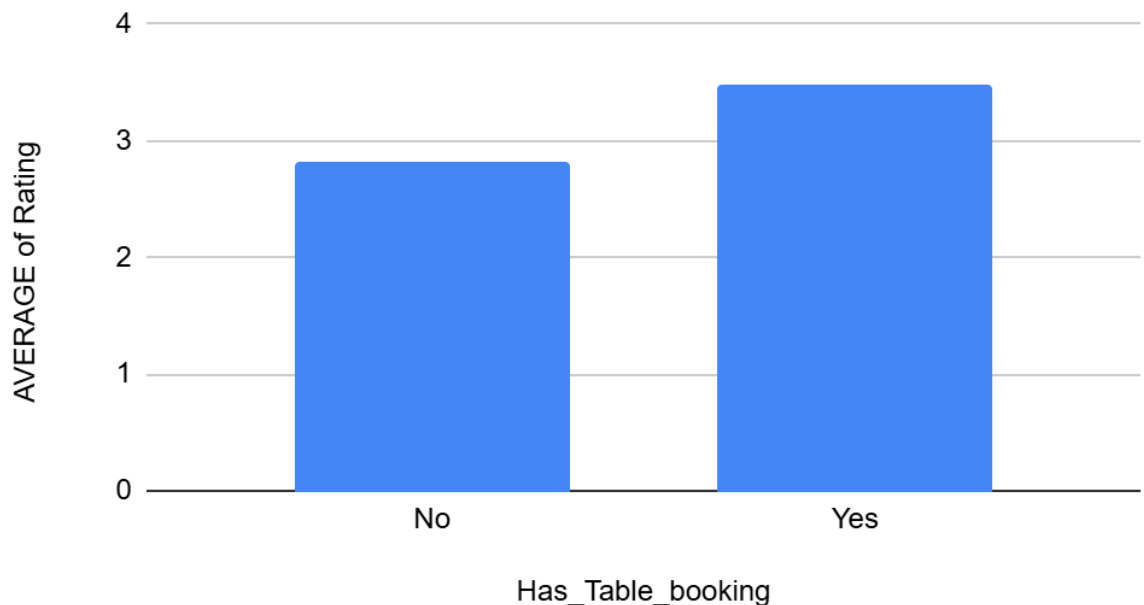
| <i>Has_Online_delivery</i> | AVERAGE of Rating |  |
|----------------------------|-------------------|--|
| No                         | 2.754309859       |  |
| Yes                        | 3.288004896       |  |

## AVERAGE of Rating



| <i>Has_Table_booking</i> | AVERAGE of Rating |
|--------------------------|-------------------|
| No                       | 2.809686644       |
| Yes                      | 3.482556131       |

## AVERAGE of Rating



### - Description:

This pivot table compares the **average customer rating** of restaurants that **offer online delivery** versus those that **do not offer online delivery**.

It helps identify whether online delivery has a positive effect on customer satisfaction.

### Table Values Used:

**No Online Delivery → 2.754**

**Yes Online Delivery → 3.288**

### Insights:

Restaurants that **offer online delivery** have an average rating of **3.288**, while restaurants without online delivery have an average rating of **2.754**.

This shows that restaurants with online delivery are rated approximately **0.53 points higher**, which indicates a **clear positive relationship** between online delivery and customer satisfaction.

Restaurants that **offer table booking** have an average rating of **3.483**, while restaurants without table booking have an average rating of **2.810**.

This means restaurants with table booking are rated approximately **0.67 points higher**, which is an even stronger positive difference compared to online delivery.

Both service features show a **positive impact on ratings**, but **table booking has a slightly stronger effect** than online delivery in the current dataset.

This suggests that customers value:

**convenience and accessibility** through online delivery

**comfort, planning, and smoother dining experience** through table booking

### **Recommendations:**

**Yes, the business should strongly consider offering both online delivery and table booking**, because both features are associated with **higher customer ratings**.

#### **Online Delivery Recommendation:**

Since restaurants with online delivery have better ratings, this feature should be included to improve:

customer convenience

reach

repeat orders

overall satisfaction

#### **Table Booking Recommendation:**

Since table booking has the **highest positive impact on ratings**, it should be treated as a **priority feature**, especially for dine-in focused outlets.

It helps improve:

customer planning

waiting-time management

dining experience

premium brand perception

#### **Strategic Recommendation:**

If the restaurant expansion model supports both dine-in and delivery, then the ideal strategy is to **implement both services together**.

If only one feature can be prioritized initially, then **table booking should be preferred first**, because it shows a **slightly stronger positive effect on ratings** in the current data.

## Question 8: Should the team keep the rate of cuisines higher? Will that affect the feedback? According to our data are the rates of cuisines and ratings, correlated?

### Solution:

#### Approach:

#### Methodology:

To analyze whether increasing the rate (price) of cuisines affects customer feedback, the dataset was systematically processed using analytical techniques. Restaurants were categorized into different **price ranges based on the average cost for two**, and for each category, the **average rating** was computed using pivot table analysis. This enabled comparison of customer feedback across various pricing levels. Further, to measure the strength and direction of the relationship between pricing and ratings, a **correlation analysis** was performed between the **average cost of cuisines** and their **corresponding ratings**. The correlation coefficient helps in identifying whether higher pricing leads to better customer satisfaction or not.

#### Assumptions:

- The dataset is assumed to be **accurate and representative** of real-world restaurant data.
- Ratings provided by customers are considered a **reliable indicator of customer satisfaction**.
- The **average cost for two** is used as a proxy for cuisine pricing.
- External factors such as location, service quality, and ambience are assumed to have **uniform influence** across the dataset for this analysis.

#### Charts & Tables:

- Table / Chart Name:

| Average cost | Average of Rating |
|--------------|-------------------|
| ₹0.00        | 3.76              |
| ₹50.00       | 1.85              |
| ₹82.50       | 4.90              |
| ₹96.25       | 4.40              |
| ₹100.00      | 1.97              |
| ₹110.00      | 4.67              |
| ₹120.00      | 2.53              |
| ₹137.50      | 4.20              |
| ₹150.00      | 2.28              |

|         |      |
|---------|------|
| ₹151.25 | 4.00 |
| ₹165.00 | 4.35 |
| ₹192.50 | 4.32 |
| ₹200.00 | 2.09 |
| ₹206.25 | 4.20 |
| ₹220.00 | 4.14 |
| ₹247.50 | 4.30 |
| ₹250.00 | 2.25 |
| ₹260.00 | 1.00 |
| ₹275.00 | 4.30 |
| ₹280.00 | 4.13 |
| ₹288.75 | 4.20 |
| ₹290.00 | 3.40 |
| ₹300.00 | 2.25 |
| ₹330.00 | 4.00 |
| ₹350.00 | 2.51 |
| ₹357.50 | 4.90 |
| ₹364.00 | 4.50 |
| ₹371.00 | 3.70 |
| ₹386.40 | 3.73 |
| ₹400.00 | 2.64 |
| ₹412.50 | 4.40 |
| ₹420.00 | 3.50 |
| ₹450.00 | 2.75 |
| ₹467.50 | 3.70 |
| ₹489.50 | 4.50 |
| ₹500.00 | 2.75 |
| ₹504.00 | 2.40 |
| ₹504.00 | 2.40 |
| ₹530.00 | 3.70 |
| ₹550.00 | 3.04 |
| ₹556.25 | 4.40 |
| ₹560.00 | 3.40 |
| ₹600.00 | 2.92 |
| ₹636.00 | 4.40 |
| ₹650.00 | 3.26 |
| ₹667.50 | 4.40 |
| ₹672.00 | 4.20 |
| ₹672.00 | 4.20 |
| ₹700.00 | 3.11 |
| ₹700.00 | 3.90 |
| ₹720.00 | 3.60 |
| ₹750.00 | 3.32 |
| ₹756.00 | 3.70 |
| ₹795.00 | 4.20 |

|           |      |
|-----------|------|
| ₹800.00   | 3.24 |
| ₹801.00   | 4.20 |
| ₹832.00   | 4.02 |
| ₹840.00   | 2.60 |
| ₹840.00   | 4.20 |
| ₹850.00   | 3.43 |
| ₹874.50   | 4.60 |
| ₹888.00   | 4.50 |
| ₹890.00   | 4.03 |
| ₹892.00   | 4.10 |
| ₹900.00   | 3.40 |
| ₹924.00   | 2.00 |
| ₹950.00   | 3.56 |
| ₹980.00   | 3.80 |
| ₹1,000.00 | 3.35 |
| ₹1,008.00 | 4.23 |
| ₹1,023.50 | 4.80 |
| ₹1,026.00 | 4.70 |
| ₹1,036.00 | 4.30 |
| ₹1,050.00 | 3.90 |
| ₹1,060.00 | 4.60 |
| ₹1,065.00 | 2.80 |
| ₹1,092.00 | 1.00 |
| ₹1,100.00 | 3.52 |
| ₹1,104.00 | 3.70 |
| ₹1,112.50 | 3.99 |
| ₹1,115.00 | 4.00 |
| ₹1,120.00 | 4.45 |
| ₹1,140.00 | 4.30 |
| ₹1,150.00 | 3.55 |
| ₹1,176.00 | 2.90 |
| ₹1,184.00 | 4.43 |
| ₹1,200.00 | 3.59 |
| ₹1,201.50 | 4.10 |
| ₹1,230.00 | 4.20 |
| ₹1,250.00 | 3.55 |
| ₹1,258.00 | 4.00 |
| ₹1,260.00 | 3.90 |
| ₹1,260.00 | 4.00 |
| ₹1,268.25 | 4.50 |
| ₹1,282.50 | 4.70 |
| ₹1,300.00 | 3.68 |
| ₹1,308.30 | 3.70 |
| ₹1,325.00 | 4.10 |

|           |      |
|-----------|------|
| ₹1,332.00 | 4.20 |
| ₹1,335.00 | 4.18 |
| ₹1,338.00 | 4.20 |
| ₹1,344.00 | 3.70 |
| ₹1,350.00 | 3.65 |
| ₹1,368.00 | 4.00 |
| ₹1,400.00 | 3.79 |
| ₹1,424.00 | 4.27 |
| ₹1,428.00 | 4.60 |
| ₹1,449.50 | 4.30 |
| ₹1,450.00 | 3.50 |
| ₹1,468.50 | 4.30 |
| ₹1,480.00 | 4.30 |
| ₹1,500.00 | 3.70 |
| ₹1,512.00 | 4.28 |
| ₹1,525.00 | 3.33 |
| ₹1,550.00 | 3.75 |
| ₹1,557.50 | 4.18 |
| ₹1,561.00 | 4.03 |
| ₹1,590.00 | 4.00 |
| ₹1,597.50 | 3.58 |
| ₹1,600.00 | 3.89 |
| ₹1,602.00 | 3.87 |
| ₹1,628.00 | 4.80 |
| ₹1,650.00 | 3.85 |
| ₹1,656.00 | 3.70 |
| ₹1,680.00 | 4.17 |
| ₹1,700.00 | 3.80 |
| ₹1,735.50 | 3.90 |
| ₹1,750.00 | 3.80 |
| ₹1,776.00 | 4.50 |
| ₹1,780.00 | 4.16 |
| ₹1,784.00 | 3.63 |
| ₹1,795.50 | 4.30 |
| ₹1,800.00 | 3.76 |
| ₹1,824.00 | 3.60 |
| ₹1,824.50 | 4.00 |
| ₹1,845.00 | 3.00 |
| ₹1,850.00 | 3.63 |
| ₹1,855.00 | 4.30 |
| ₹1,895.50 | 3.90 |
| ₹1,900.00 | 3.65 |
| ₹1,950.00 | 4.15 |
| ₹2,000.00 | 3.55 |

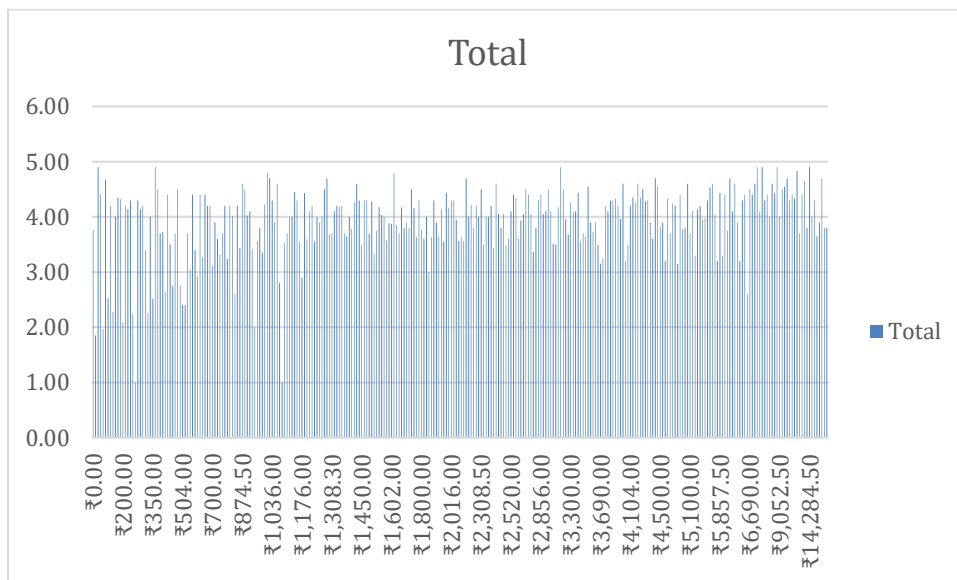


|           |      |
|-----------|------|
| ₹2,002.50 | 4.44 |
| ₹2,007.00 | 4.17 |
| ₹2,016.00 | 4.30 |
| ₹2,052.00 | 4.30 |
| ₹2,080.00 | 3.95 |
| ₹2,100.00 | 3.56 |
| ₹2,130.00 | 3.63 |
| ₹2,200.00 | 3.56 |
| ₹2,220.00 | 4.70 |
| ₹2,225.00 | 4.00 |
| ₹2,230.00 | 4.22 |
| ₹2,280.00 | 3.80 |
| ₹2,291.75 | 4.20 |
| ₹2,300.00 | 4.00 |
| ₹2,308.50 | 4.50 |
| ₹2,350.00 | 3.50 |
| ₹2,352.00 | 4.00 |
| ₹2,380.75 | 4.00 |
| ₹2,385.00 | 4.20 |
| ₹2,400.00 | 3.44 |
| ₹2,425.25 | 4.60 |
| ₹2,453.00 | 4.05 |
| ₹2,460.00 | 3.80 |
| ₹2,496.00 | 4.05 |
| ₹2,500.00 | 3.49 |
| ₹2,508.00 | 3.60 |
| ₹2,520.00 | 4.10 |
| ₹2,536.50 | 4.40 |
| ₹2,565.00 | 4.34 |
| ₹2,600.00 | 3.60 |
| ₹2,650.00 | 3.93 |
| ₹2,662.50 | 4.05 |
| ₹2,664.00 | 4.50 |
| ₹2,670.00 | 4.40 |
| ₹2,676.00 | 4.05 |
| ₹2,700.00 | 3.37 |
| ₹2,800.00 | 3.80 |
| ₹2,821.50 | 4.30 |
| ₹2,856.00 | 4.40 |
| ₹2,899.00 | 4.05 |
| ₹2,900.00 | 4.10 |
| ₹2,912.00 | 4.50 |
| ₹2,960.00 | 4.10 |
| ₹3,000.00 | 3.51 |

|           |      |
|-----------|------|
| ₹3,075.00 | 3.50 |
| ₹3,078.00 | 4.18 |
| ₹3,115.00 | 4.90 |
| ₹3,122.00 | 4.50 |
| ₹3,195.00 | 3.95 |
| ₹3,200.00 | 3.68 |
| ₹3,300.00 | 4.25 |
| ₹3,328.00 | 4.09 |
| ₹3,334.50 | 4.10 |
| ₹3,345.00 | 4.44 |
| ₹3,360.00 | 3.55 |
| ₹3,420.00 | 3.70 |
| ₹3,500.00 | 3.64 |
| ₹3,568.00 | 4.55 |
| ₹3,591.00 | 3.90 |
| ₹3,600.00 | 3.73 |
| ₹3,648.00 | 3.90 |
| ₹3,650.00 | 3.50 |
| ₹3,690.00 | 3.15 |
| ₹3,700.00 | 3.25 |
| ₹3,727.50 | 4.19 |
| ₹3,744.00 | 4.10 |
| ₹3,791.00 | 4.30 |
| ₹3,800.00 | 4.30 |
| ₹3,864.00 | 4.33 |
| ₹3,876.00 | 4.20 |
| ₹4,000.00 | 3.96 |
| ₹4,014.00 | 4.60 |
| ₹4,032.00 | 3.20 |
| ₹4,100.00 | 3.50 |
| ₹4,104.00 | 4.20 |
| ₹4,160.00 | 4.35 |
| ₹4,200.00 | 4.26 |
| ₹4,237.00 | 4.60 |
| ₹4,240.00 | 4.35 |
| ₹4,249.75 | 4.50 |
| ₹4,260.00 | 4.28 |
| ₹4,270.00 | 4.30 |
| ₹4,300.00 | 3.90 |
| ₹4,400.00 | 3.60 |
| ₹4,440.00 | 4.70 |
| ₹4,460.00 | 4.55 |
| ₹4,500.00 | 3.82 |
| ₹4,560.00 | 3.90 |

|           |      |
|-----------|------|
| ₹4,612.50 | 3.20 |
| ₹4,617.00 | 4.33 |
| ₹4,700.00 | 3.70 |
| ₹4,792.50 | 4.24 |
| ₹4,800.00 | 4.20 |
| ₹4,920.00 | 3.15 |
| ₹4,992.00 | 4.40 |
| ₹5,000.00 | 3.78 |
| ₹5,016.00 | 3.80 |
| ₹5,040.00 | 4.60 |
| ₹5,100.00 | 3.70 |
| ₹5,129.00 | 4.10 |
| ₹5,130.00 | 3.30 |
| ₹5,325.00 | 4.13 |
| ₹5,386.50 | 4.20 |
| ₹5,500.00 | 3.95 |
| ₹5,575.00 | 3.98 |
| ₹5,643.00 | 4.30 |
| ₹5,700.00 | 4.53 |
| ₹5,798.00 | 4.60 |
| ₹5,824.00 | 4.04 |
| ₹5,842.50 | 3.20 |
| ₹5,857.50 | 4.44 |
| ₹5,880.00 | 3.30 |
| ₹5,920.00 | 4.40 |
| ₹6,000.00 | 3.75 |
| ₹6,021.00 | 4.70 |
| ₹6,150.00 | 4.10 |
| ₹6,156.00 | 4.60 |
| ₹6,244.00 | 3.90 |
| ₹6,355.50 | 3.20 |
| ₹6,390.00 | 4.30 |
| ₹6,500.00 | 4.40 |
| ₹6,624.00 | 2.60 |
| ₹6,690.00 | 4.50 |
| ₹6,720.00 | 4.40 |
| ₹6,840.00 | 4.60 |
| ₹6,853.00 | 4.90 |
| ₹7,000.00 | 4.10 |
| ₹7,359.00 | 4.90 |
| ₹7,455.00 | 4.30 |
| ₹7,805.00 | 4.40 |
| ₹8,000.00 | 4.00 |
| ₹8,320.00 | 4.60 |

|                    |             |
|--------------------|-------------|
| ₹8,520.00          | 4.43        |
| ₹8,880.00          | 4.90        |
| ₹9,052.50          | 4.00        |
| ₹9,120.00          | 4.50        |
| ₹9,585.00          | 4.55        |
| ₹9,747.00          | 4.70        |
| ₹10,146.00         | 4.30        |
| ₹10,260.00         | 4.40        |
| ₹10,650.00         | 4.33        |
| ₹11,150.00         | 4.83        |
| ₹11,400.00         | 3.70        |
| ₹12,540.00         | 4.40        |
| ₹12,780.00         | 4.65        |
| ₹13,530.00         | 3.80        |
| ₹14,284.50         | 4.90        |
| ₹16,605.00         | 4.00        |
| ₹17,040.00         | 4.30        |
| ₹18,450.00         | 3.65        |
| ₹19,372.50         | 3.90        |
| ₹24,495.00         | 4.70        |
| ₹26,445.00         | 3.80        |
| ₹30,750.00         | 3.80        |
| <b>Grand Total</b> | <b>2.89</b> |



- Description:

The chart represents the relationship between **Average Cost for Two (in INR)** on the X-axis and **Average Rating** on the Y-axis for various restaurants. Each bar corresponds to a specific cost value and its associated average rating.

From the visualization, it can be observed that the **ratings are distributed between approximately 2.0 to 4.8**, with most values concentrated around the **3.5 to 4.5**

**range.** Despite a wide variation in cost (ranging from low to very high), the ratings do not show a consistent increasing or decreasing trend.

The chart indicates that **higher-priced restaurants do not always receive higher ratings**, and similarly, some lower-cost restaurants achieve comparable or even better ratings. This suggests that **pricing is not the sole factor influencing customer satisfaction**.

Overall, the pattern appears **scattered and irregular**, which supports the earlier correlation result of a **weak positive relationship** between cost and rating.

Therefore, factors such as **food quality, service, and customer experience** are likely more significant in determining ratings than price alone.

### Insights:

- The ratings fluctuate across different price ranges, showing **no consistent trend** that higher-priced cuisines always receive better ratings.
- Mid-range cuisines often perform **better than both low and high-priced categories**, indicating balanced pricing plays an important role.
- Some low-cost cuisines also maintain **reasonable ratings**, suggesting affordability combined with quality can satisfy customers.
- The overall **average rating is around 2.89**, highlighting scope for improvement in customer satisfaction.
- The calculated **correlation coefficient is 0.2205**, which indicates a **weak positive relationship** between cuisine price and ratings.
- This weak correlation shows that **pricing has only a minor impact**, and other factors significantly influence customer feedback.

### Recommendations:

- The team should **not focus solely on increasing cuisine prices**, as it does not guarantee improved ratings.
- Greater emphasis should be placed on **enhancing food quality, service, and overall customer experience**.
- Maintain a **balanced and competitive pricing strategy** to attract a wider customer base.
- Focus on delivering **value for money**, especially in mid-range pricing where customer satisfaction is relatively higher.
- Improve **digital services such as online delivery and table booking**, as they positively influence ratings.
- Continuously monitor customer feedback to identify **key drivers beyond pricing** for better decision-making.

### Conclusion (Correlation Interpretation):

The correlation value of **0.2205** indicates a **weak positive relationship** between cuisine price and customer ratings. This implies that increasing prices has only a **limited impact on customer feedback**. Therefore, it can be concluded that **higher pricing alone will not significantly improve ratings**, and the focus should instead be on **quality, service, and customer satisfaction** to achieve better outcomes.

## Question 9: What is the distribution of the number of restaurants of different price ranges in all the countries?

### Solution:

#### Approach:

To understand the **distribution of restaurants across different price ranges in all countries**, I analyzed the number of restaurants in each **Price Range category (1 to 4)** country-wise.

#### Methodology used:

Used **Country** as the row field in the Pivot Table.

Used **Price\_range (1, 2, 3, 4)** as the column field.

Used **COUNTA of RestaurantID** as the value field to count the number of restaurants in each price range.

This allowed a country-wise comparison of how restaurants are distributed from:

**Low-price segment (Price Range 1)**

**to Premium segment (Price Range 4)**

#### Assumption:

**Price Range 1** = Low-cost / budget restaurants

**Price Range 2** = Lower-mid segment

**Price Range 3** = Mid-to-premium segment

**Price Range 4** = Premium / luxury segment

This analysis helps identify:

which countries are dominated by budget restaurants

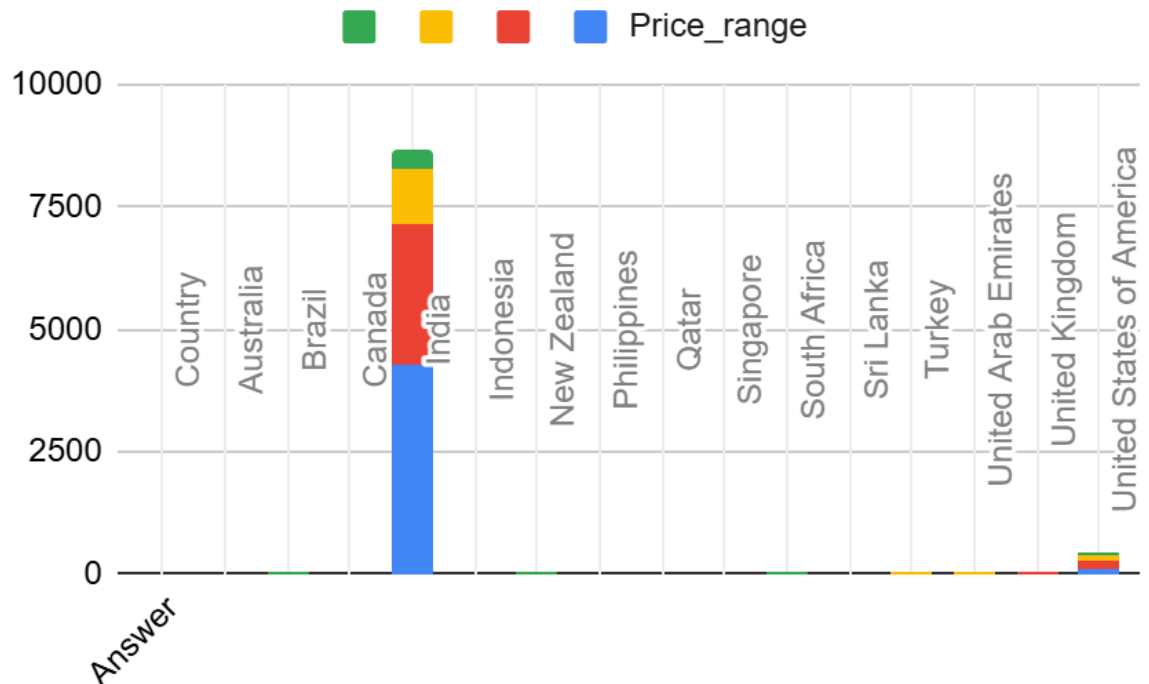
which countries have stronger premium dining presence

where the market is more balanced across all price categories

#### Charts & Tables:

- Table / Chart Name:

|               | <i>COUNTA of<br/>RestaurantID</i> | <i>Price_range</i> |          |          |          |
|---------------|-----------------------------------|--------------------|----------|----------|----------|
| <b>Answer</b> | <b>Country</b>                    | <b>1</b>           | <b>2</b> | <b>3</b> | <b>4</b> |
|               | Australia                         | 4                  | 14       | 5        | 1        |
|               | Brazil                            | 2                  | 7        | 16       | 35       |
|               | Canada                            |                    | 3        |          | 1        |
|               | India                             | 4295               | 2858     | 1111     | 388      |
|               | <b>Indonesia</b>                  |                    | 1        | 20       |          |
|               | New Zealand                       | 3                  | 4        | 17       | 16       |
|               | Philippines                       |                    | 1        | 12       | 9        |
|               | Qatar                             |                    | 1        | 5        | 14       |
|               | Singapore                         |                    | 1        | 5        | 14       |
|               | South Africa                      |                    | 4        | 17       | 39       |
|               | Sri Lanka                         |                    | 6        | 11       | 3        |
|               | Turkey                            |                    | 11       | 18       | 5        |
|               | United Arab<br>Emirates           |                    | 9        | 29       | 22       |
|               | United<br>Kingdom                 | 4                  | 28       | 32       | 16       |
|               | United States<br>of America       | 136                | 165      | 110      | 23       |



- Description:

This pivot table shows the **number of restaurants in each price range category (1–4)** for every country.

It helps understand the **market structure** of each country by showing whether restaurants are concentrated in budget, mid-range, or premium segments.

### Insights:

#### 1. India has the largest restaurant distribution by far

India has the **highest number of restaurants in every price range**, especially:

Price Range 1 → 4295

Price Range 2 → 2858

This shows that India is a **massive market dominated by budget and affordable dining**.

Even though India also has restaurants in higher price ranges, the market is heavily concentrated in **low to mid-price segments**.

#### 2. India is strongly budget-driven



In India, the highest concentration is in:

**Price Range 1**

**Price Range 2**

This means the Indian market is strongly focused on:

affordable pricing

large customer volume

value-for-money dining

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### **3. Several countries are premium-heavy**

Some countries have a stronger concentration in **Price Range 3 and 4**, meaning they are more premium-oriented:

**Brazil** → Strong in **Price Range 4 (35)**

**Qatar** → Strong in **Price Range 4 (14)**

**Singapore** → Strong in **Price Range 4 (14)**

**South Africa** → Very strong in **Price Range 4 (39)**

**United Arab Emirates** → Strong in **Price Range 3 (29)** and **4 (22)**

**New Zealand** → Balanced but strong in **Price Range 3 (17)** and **4 (16)**

This suggests these countries have stronger demand for:

premium dining

upscale restaurants

higher-spending customers

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### **4. Countries like Indonesia and Philippines are concentrated in mid-to-premium ranges**

**Indonesia:**

Mostly **Price Range 3 (20)**

**Philippines:**

Mostly **Price Range 3 (12)** and **Price Range 4 (9)**

This indicates that these markets may be more favorable for:

quality-focused mid-to-premium restaurant models

less dependence on low-budget mass market dining

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## **5. USA and UK show more diversified distribution**

**United States of America:**

Spread across all ranges, but strongest in **Price Range 2 (165)** and **Price Range 1 (136)**

**United Kingdom:**

Strong presence in **Price Range 2 (28)** and **Price Range 3 (32)**

This suggests:

these are **more balanced and mature markets**

multiple customer segments exist across budget to premium categories

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## **6. Some countries have limited restaurant count in the dataset**

Countries like:

**Canada**

**Australia**

**Indonesia**

**Philippines**

**Qatar**

**Singapore**

have relatively smaller counts in the dataset, so while the pattern is useful, conclusions for these countries should be interpreted with some caution compared to larger markets like India or USA.

### **Recommendations:**

#### **1. Match expansion strategy with each country's dominant price range**

The business should **not use one pricing model for all countries**.

Instead, it should align expansion based on each country's restaurant distribution pattern.

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#### **2. For India, prioritize budget and affordable models**

Since India is heavily concentrated in **Price Range 1 and 2**, the best strategy is:

budget-friendly pricing

high-volume operations

strong value-for-money positioning

**Recommended model for India:**

**Affordable / mass-market restaurant concept**

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#### **3. For premium-heavy countries, focus on mid-to-premium concepts**

Countries like:

**Qatar**

**Singapore**

**South Africa**

**Brazil**

**UAE**

**New Zealand**

show stronger presence in **Price Range 3 and 4**, so these are better suited for:

premium dining

higher-quality experience

upscale branding

**Recommended model:**

**Mid-to-premium / premium restaurant concept**

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#### **4. For balanced markets, use flexible segmentation**

Countries like:

**USA**

**UK**

have restaurants across multiple price ranges, so the team can:

choose the target segment based on location and customer type

test mid-range models first

then expand into premium or budget clusters

**Recommended model:**

**Segment-based expansion strategy**

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#### **5. For countries with small dataset counts, validate before final expansion**

For countries with limited restaurant counts in the data:

use this as an initial directional insight only

conduct further market validation before making major investment decisions

**Question 10: Explain your approach in brief for suggesting countries/cities in order to open new restaurants, if the objective and subjective questions would have not been given to assist you. [you have to give bullet pointers in order to answer this question]**

## **Solution:**

### **Approach:**

#### **Methodology**

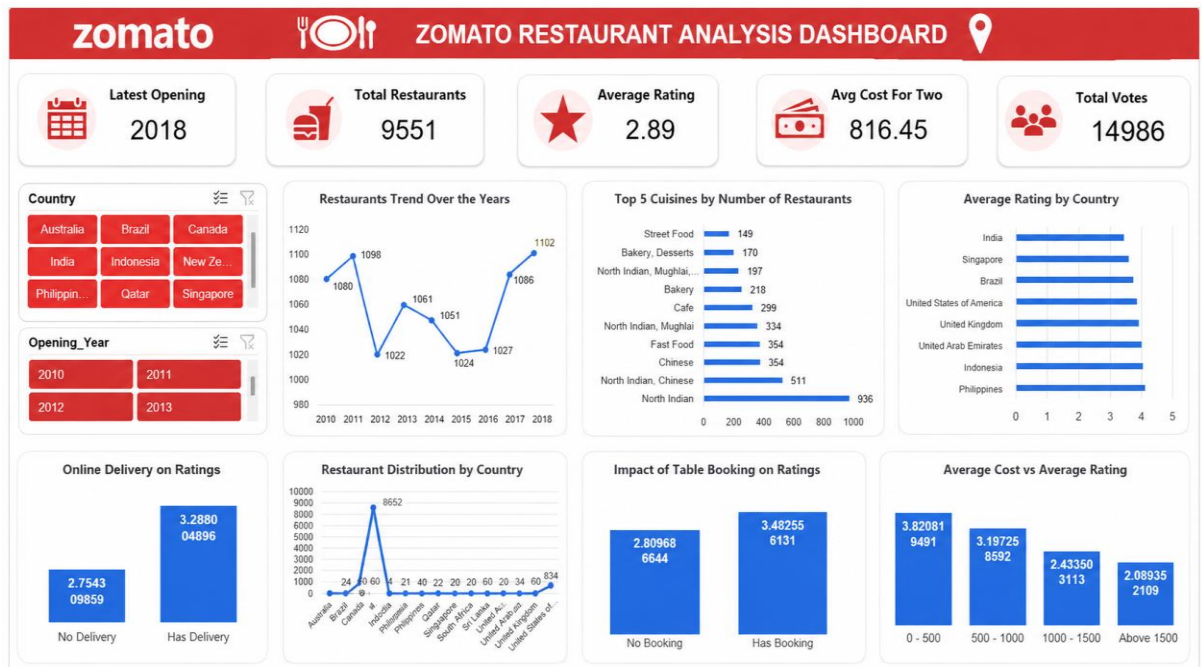
- Analyzed **restaurant distribution across countries** to identify regions with **lower competition**.
- Evaluated **average ratings** to understand customer satisfaction levels in each country and city.
- Compared **average cost for two** to identify affordability and value-for-money locations.
- Studied **customer engagement (votes)** to measure demand and popularity.
- Used **pivot tables and charts** to filter and shortlist high-potential countries and cities.
- Performed **multi-factor analysis (competition + rating + cost + demand)** to ensure balanced decision-making.

#### **Assumptions**

- Dataset is assumed to be **accurate and representative** of actual market conditions.
- **Ratings reflect customer satisfaction** reliably.
- **Average cost for two** represents pricing level of the market.
- External factors like **location, service quality, and brand influence** are considered uniform for analysis.

### **Charts & Tables:**

- Table / Chart Name:



#### - Description:

The **Zomato Restaurant Analysis Dashboard** provides a comprehensive overview of restaurant data across multiple countries, helping in understanding trends, customer behavior, and business performance.

The dashboard highlights key performance indicators such as **total number of restaurants (9551)**, **average rating (2.89)**, **average cost for two (₹816.45)**, and **total votes (14,986)**. It also shows the **latest opening year (2018)**, indicating recent platform growth.

Interactive filters for **country** and **opening year** allow users to dynamically explore data and gain customized insights.

The dashboard includes multiple visualizations:

- **Trend Analysis:** Shows how restaurant openings have changed over the years, indicating overall growth with slight fluctuations.
- **Cuisine Analysis:** Identifies the most popular cuisines, with **North Indian cuisine dominating**.
- **Geographical Insights:** Displays restaurant distribution and average ratings across countries, highlighting that **India has the highest number of restaurants**.
- **Customer Behavior Insights:** Compares ratings based on **online delivery and table booking**, showing that both services positively impact ratings.
- **Cost vs Rating Analysis:** Helps understand how pricing affects customer satisfaction.

Overall, the dashboard enables **data-driven decision-making** by combining operational metrics, customer preferences, and market trends in a single, visually intuitive interface.

#### Insights:

- Countries with **lower restaurant count** indicate **less competition and higher opportunity** for expansion.

- Regions like **Canada, Qatar, Singapore, Sri Lanka, Indonesia, and Philippines** show a balance of **moderate competition and good ratings**.
- Some cities within these countries have **high ratings but fewer restaurants**, indicating **untapped demand**.
- Customer engagement (votes) shows that **active markets exist even outside highly saturated countries**.
- Pricing varies across regions, but **mid-range cost locations often perform well in ratings**, suggesting better customer satisfaction.

**Recommendations:**

- ☐ Focus on countries with **low competition and good average ratings** for expansion.
- ☐ Target cities where **restaurant count is low but ratings are high**, indicating growth potential.
- ☐ Maintain a **balanced pricing strategy** to attract customers while ensuring profitability.
- ☐ Invest in **online delivery and table booking services**, as they positively impact ratings.
- ☐ Avoid highly saturated markets and prioritize **emerging markets with rising demand**.